



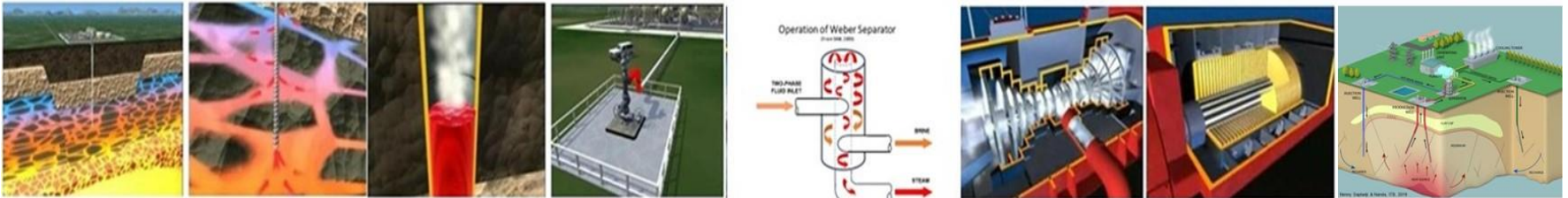
TRAINING:
SISTEM, TEKNOLOGI DAN EVALUASI FEASIBILITY STUDY GEOTERMAL

UTILISASI ENERGI PANAS BUMI

Ir. Nenny Miryani Saptadji Ph.D

Bandung, Senin 2 Desember 2024

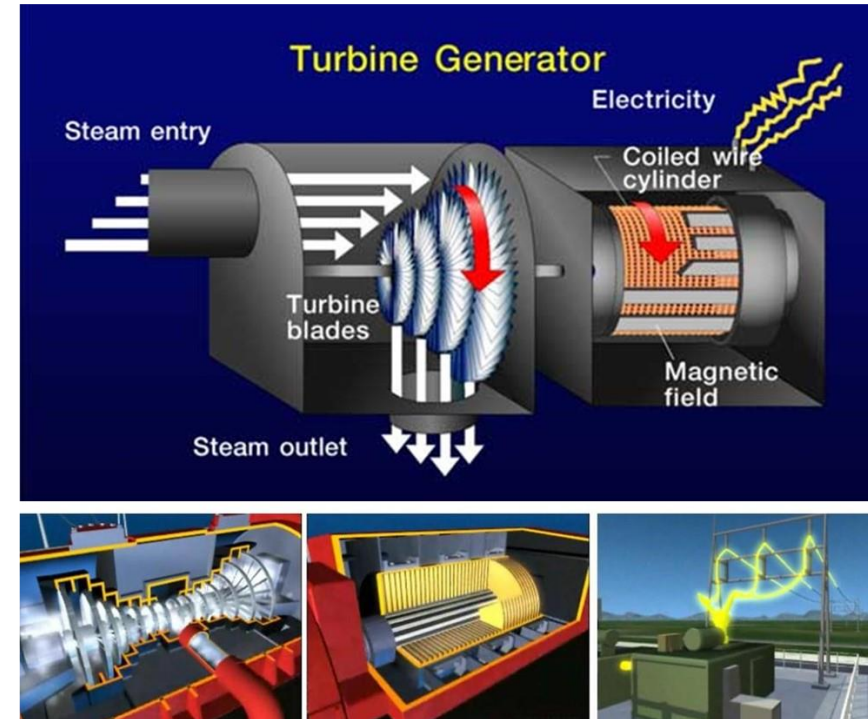
Waktu: 13.00 – 15.00



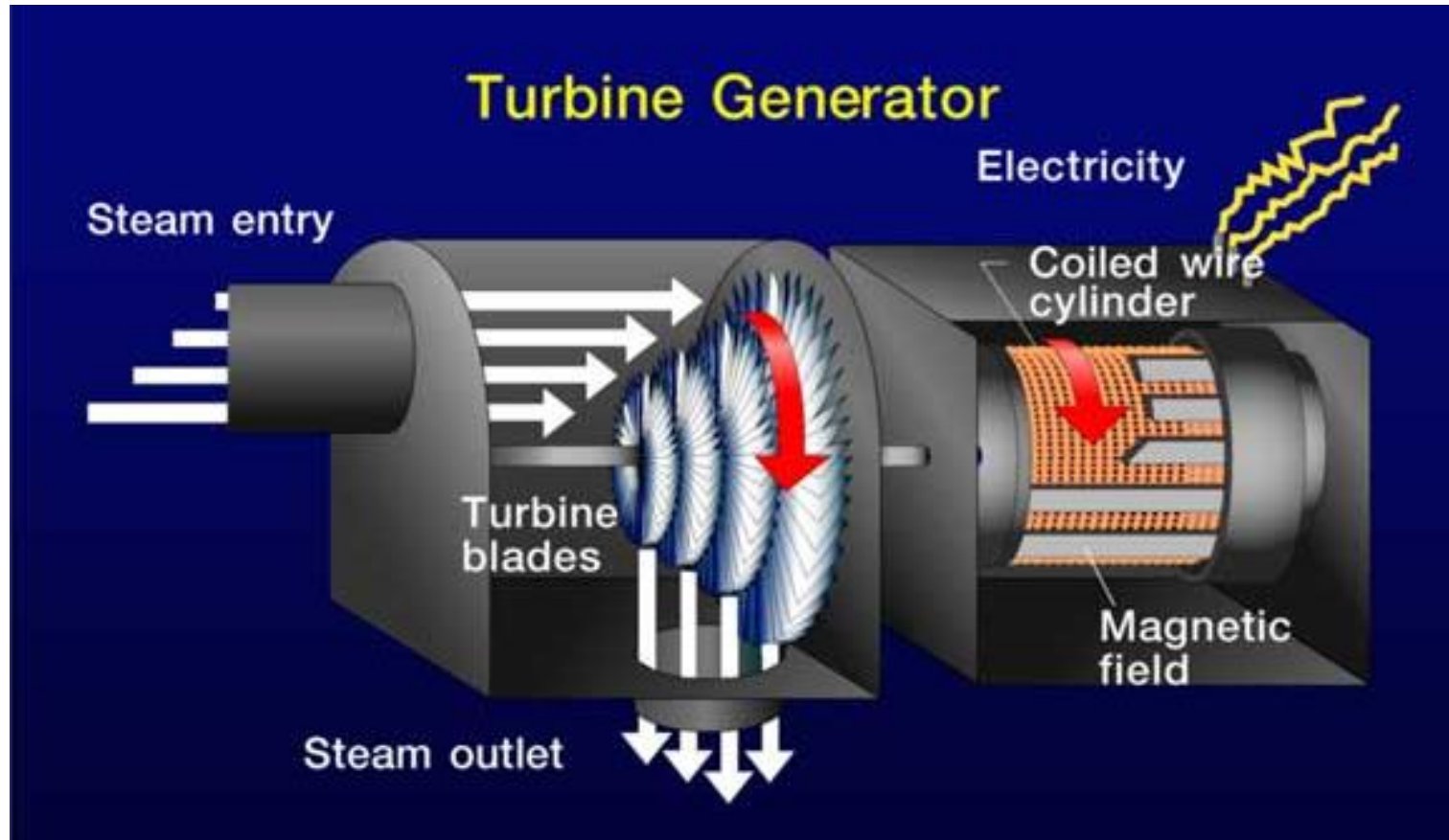
NON-ELECTRICITY UTILIZATION OF GEOTHERMAL ENERGY (DIRECT USE)

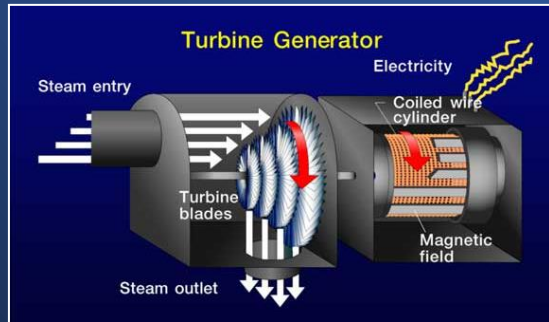


GEOHERMAL ENERGY FOR ELECTRICITY GENERATION



FLUIDA GEOTERMAL UNTUK PEMANFAATAN TIDAK LANGSUNG: memanfaatkan energi panas bumi untuk pembangkit listrik

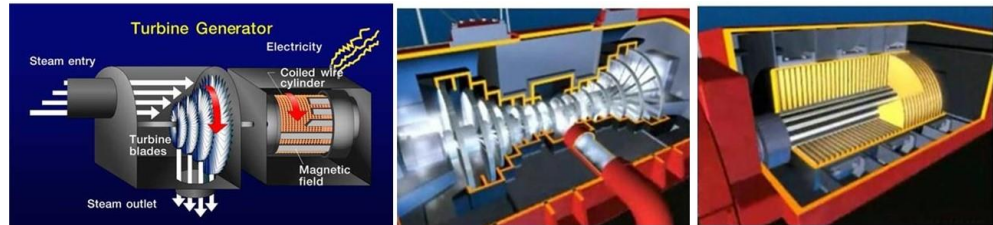




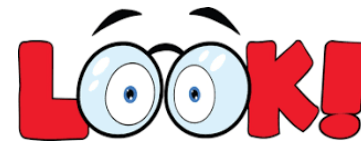
Turbin mengubah energi kinetik (dari kecepatan fluida) menjadi energi mekanik (putaran dan daya)

Jenis-jenis siklus konversi energi

1. *Direct Dry Steam*
2. *Separated Steam*
3. *Single Flash Steam*
4. *Double Flash Steam*
5. *Triple Flash Cycle*
6. *Brine Cycle*
7. *Combined Cycle*
8. *Well Head Generating Unit*

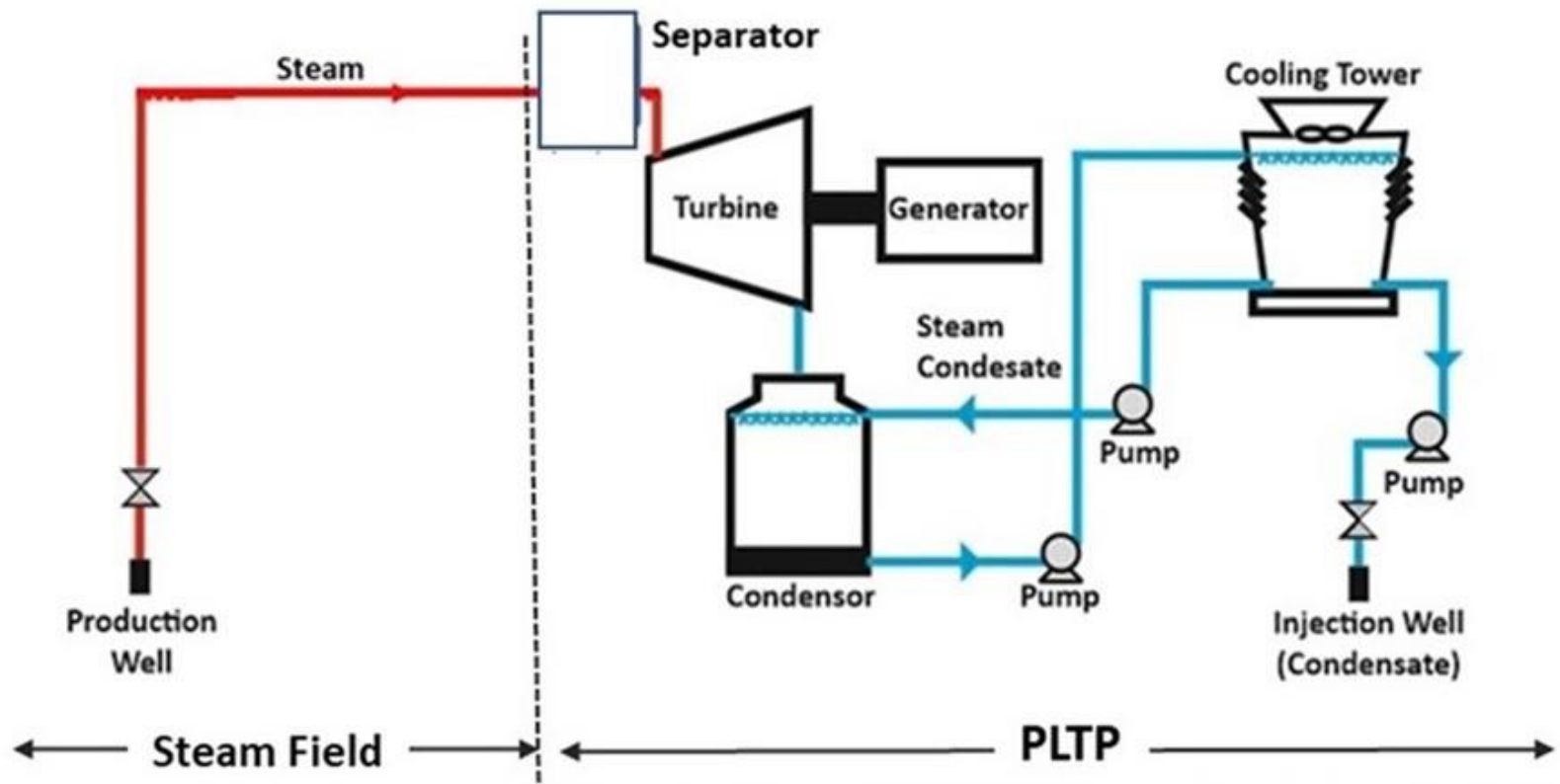


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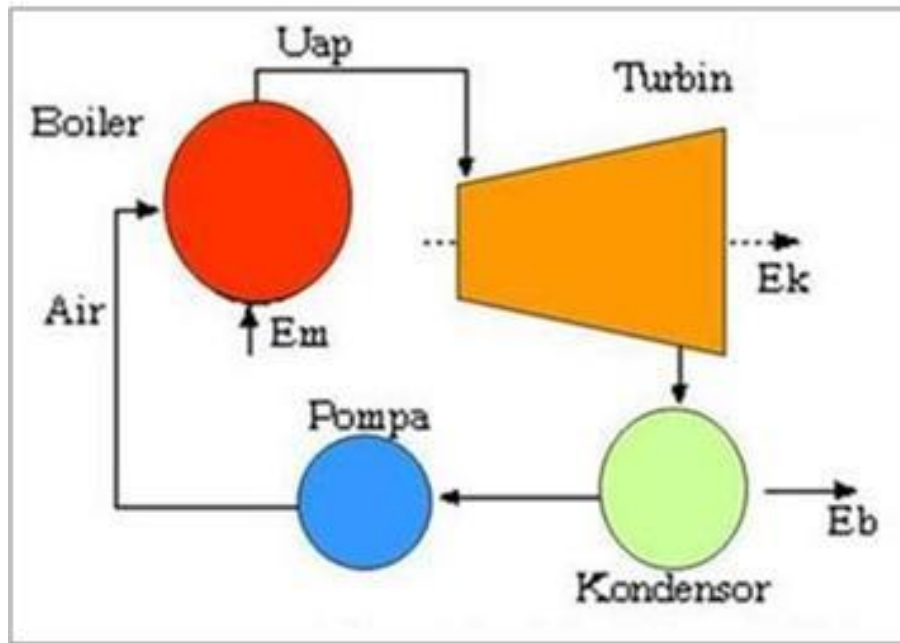
1. Jenis-jenis siklus konversi energi yang digunakan di pembangkit listrik panas bumi (PLTP).
2. Komponen PLTP
3. Jumlah uap yang dibutuhkan oleh PLTP

PLTP SIKLUS UAP KERING (*DIRECT DRY STEAM*)

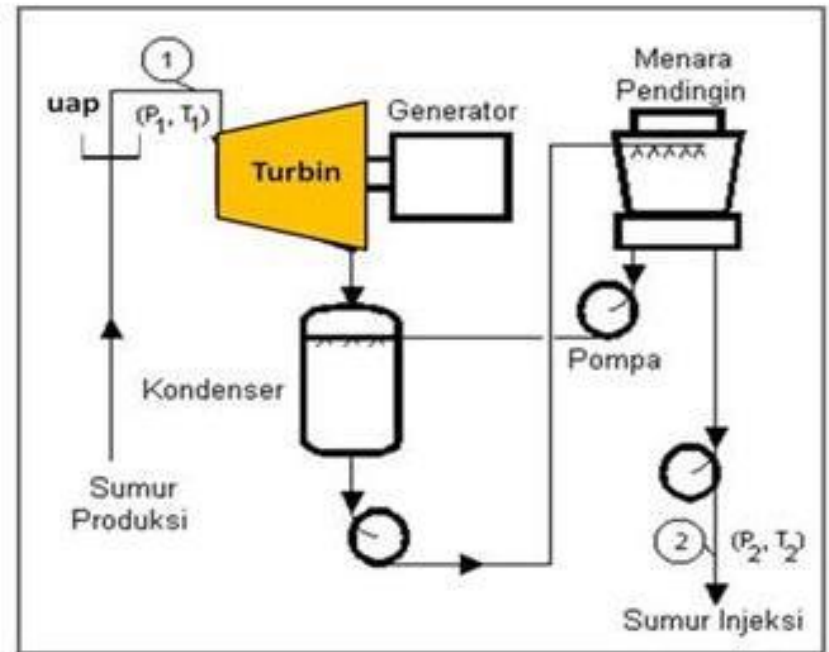


PERBANDINGAN PLTU DAN PLTP SIKLUS UAP KERING (DIRECT DRY STEAM)

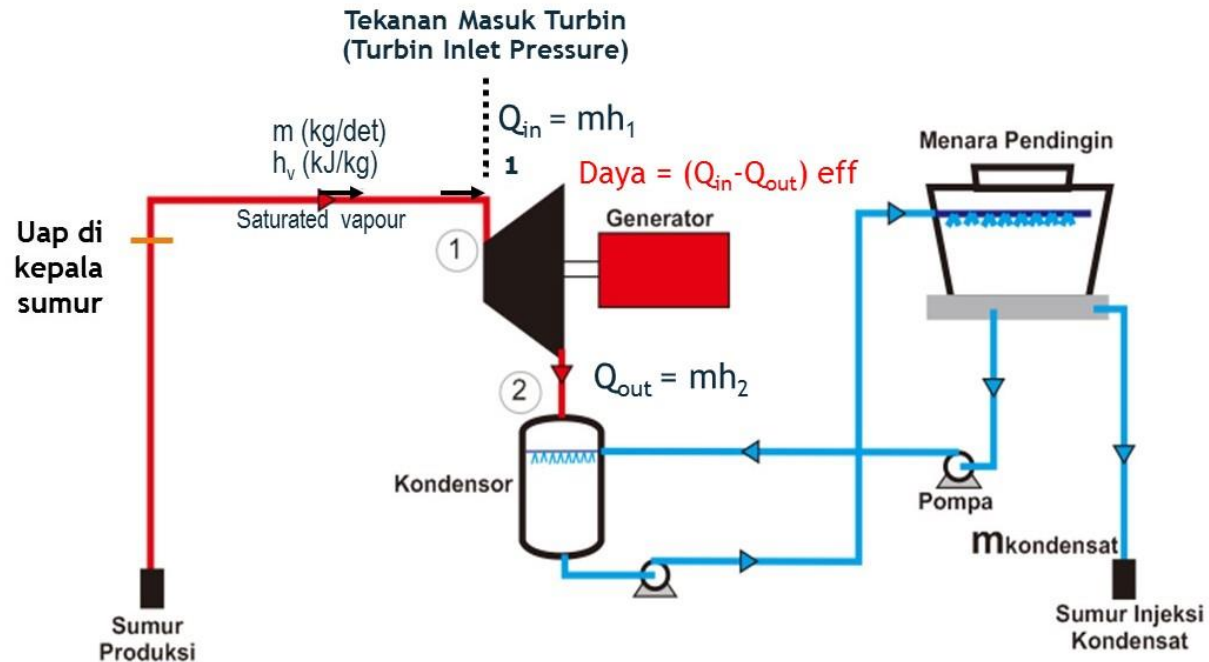
PLTU



PLTP – SIKLUS UAP KERING



METODA PENENTUAN KEBUTUHAN UAP UNTUK PLTP SIKLUS UAP KERING (DIRECT DRY STEAM)



Untuk mensuplai PLTP dengan kapasitas W Mwe, maka uap yang dibutuhkan adalah:

$$m = \frac{W}{\eta(h_1 - h_2)}$$

LATIHAN: PLTP KAMOJANG



Sumur-sumur di lapangan panasbumi Kamojang umumnya menghasilkan uap jenuh (saturated steam). Kapasitas listrik terpasang di PLTP Kamojang adalah 235 MWe.

Unit-1 30 MWe (beroperasi sejak 1983)

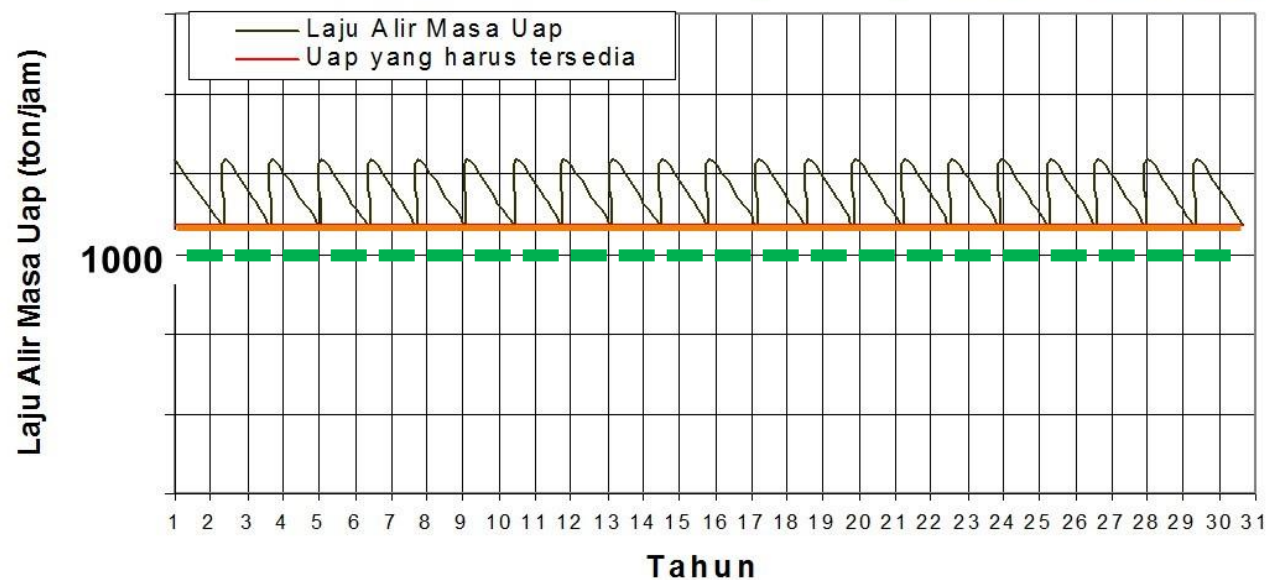
- Tekanan masuk turbin = 5.5 barg
- Tekanan condensor= 0.10 bara
- Efisiensi turbin = 80 %

Berapakah banyaknya uap yang dibutuhkan (dalam ton/jam) ?

Lakukan perhitungan untuk Unit-2 dan Unit-3 yang masing-masing mempunyai kapasitas 55 MW (beroperasi sejak 1987)

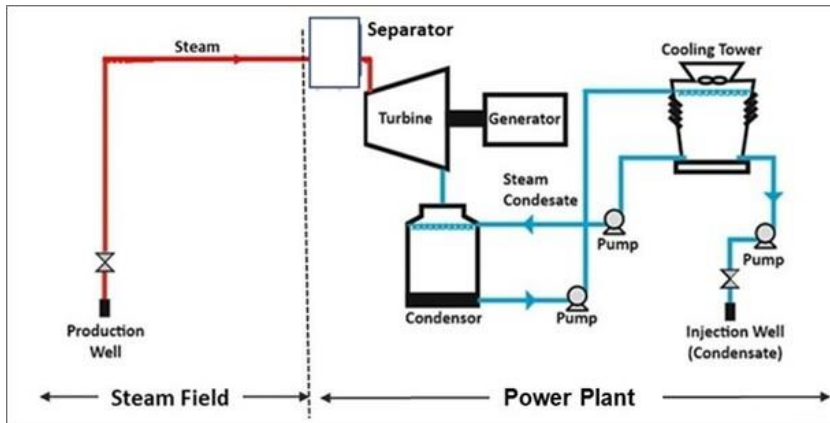
Jumlah uap yang dibutuhkan
oleh PLTP Kamojang 140 MW
Bila steam allowance-nya 10% =
1100 ton/jam

Hasil perhitungan oleh Nenny Saptadji/ITB



GEOTHERMAL ENERGY PRODUCTION SYSTEM FOR ELECTRICITY GENERATION

First developed in Indonesia is 'Direct Dry Steam Cycle'



Field Name	Installed Capacity (MW)	Total (MW)
Kamojang	1 x 30 (1983) 2 x 55 (1987) 1 x 60 (2007) 1 x 35 (2015)	235
Darajat	1 x 55 (1994) 1 x 94 (2000) 1 x 121 (2007)	270
Patuha	1 x 55 (2014)	55



Steam transmission line
(from wells to power plant)



Steam receiving
header (near the
gate to power plant)

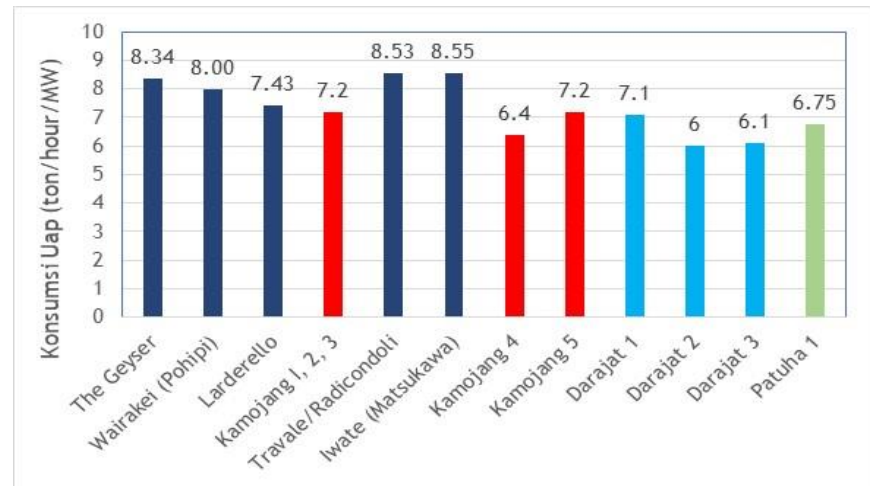


Separator



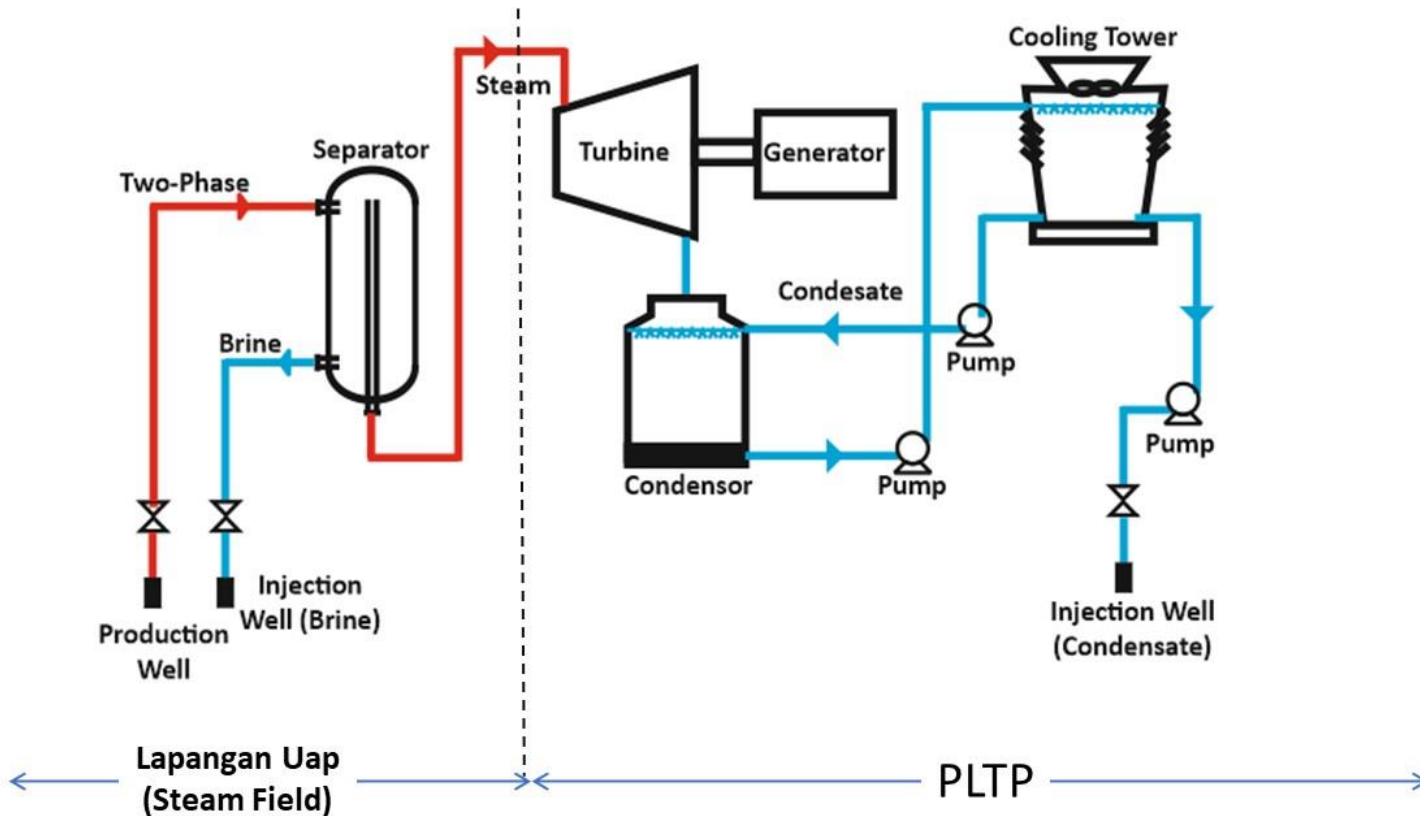
Turbine &
Condensor

PERFORMANCE OF DIRECT DRY STEAM CYCLE POWER PLANT IN INDONESIA



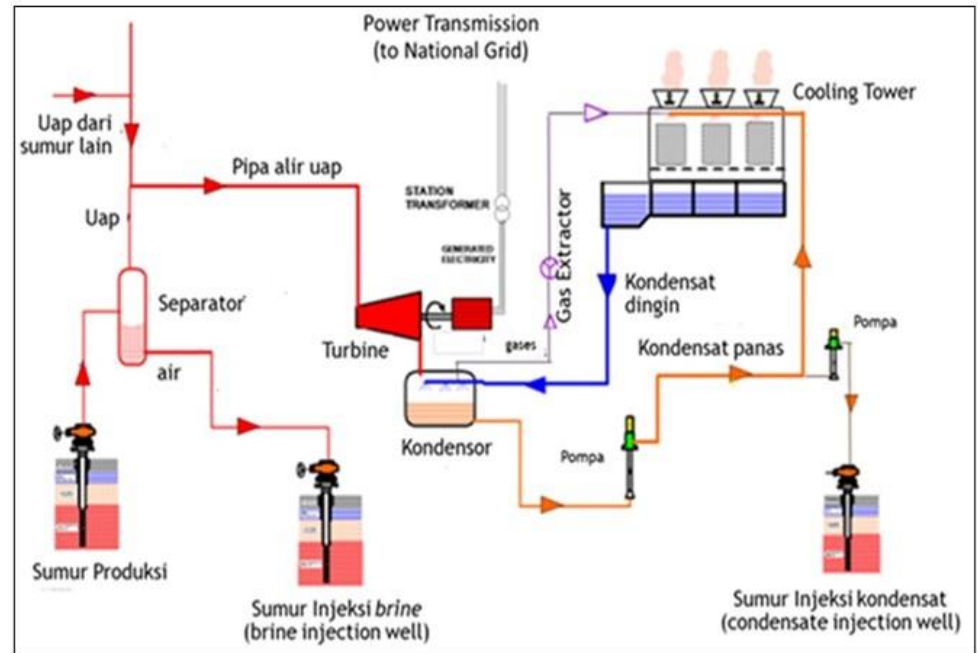
- Geothermal Power Plant (GPP) has baseload capability.
- Efficient. Steam consumption 6 – 7.2 ton/h`our/MW)
- Capacity Factor > 90%.e.g
 - Darajat Power Plant: *Capacity Factor* 92,15%, *Realibility Factor* 99,66
- Geothermal energy is a reliable energy for electricity generation, as it can generate electricity 24 hours in a day, 7 days in a week, 365 days in a year
- Kamojang Field has proven that geothermal energy is a sustainable energy resource. With proper management (good reservoir management), after 40 years the power plant is still operating properly.
- Proven project lifetime, e.i Kamojang 40 years, Darajat 30 years

SKEMA DIAGRAM PEMBANGKIT LISTRIK *SEPARATED STEAM CYCLE*.

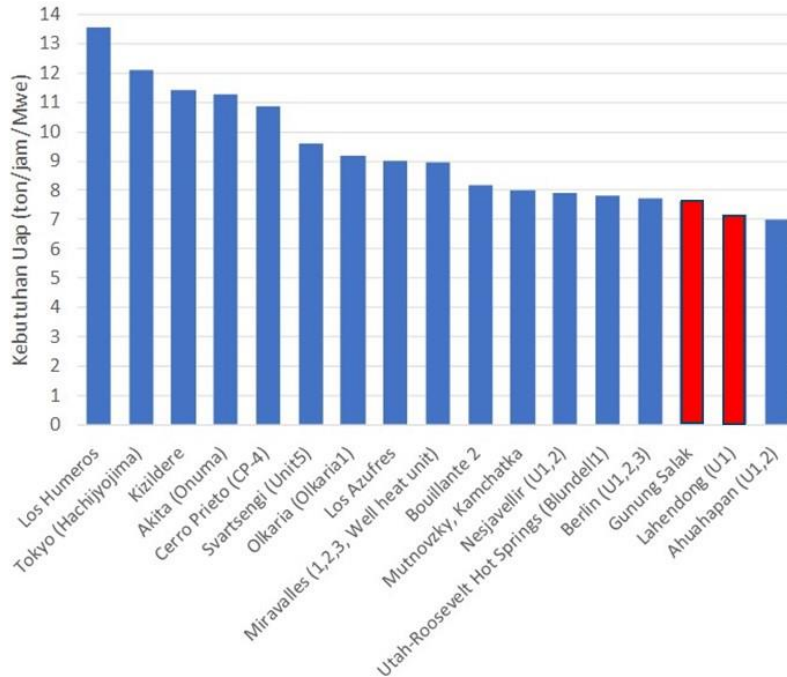


SEPARATED CYCLE (SIKLUS UAP HASIL PEMISAHAN)

Lapangan	Kapasitas (MW)	Total (MW)
Salak	3 x 60 (1994) 3 x 65,6	377
Wayang Windu	1 x 110 (2000) 1 x 117 (2009)	227
Dieng	1 x 60	60
Lahendong	4 x 20	80
Ulubelu	3 x 55	165
Ulumbu	4 x 2,5	10
Mataloko	1 x 2,5	2,5
Muara Laboh	85	85
Lumut Balai	55	55
Rantau Dedap	98,4	98,4



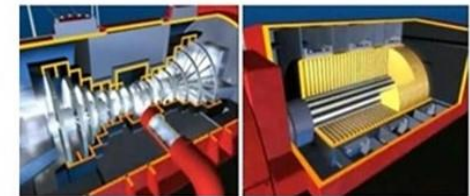
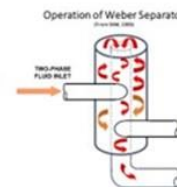
PERFORMANCE OF SEPARATED STEAM CYCLE POWER PLANT IN INDONESIA



- The Power Plant has baseload capability
- Efficient. Low steam consumption
- Capacity Factor > 90%
- Geothermal energy is a reliable energy for electricity generation, as it can generate electricity 24 hours in a day, 7 days in a week, 365 days in a year
- Proven project life time, Salak: 30 years

Wayang Windu Power Plant

	FAKTOR KETERSEDIAAN (%) AVAILABILITY FACTOR	FAKTOR KEANDALAN (%) RELIABILITY FACTOR	FAKTOR KAPASITAS (%) CAPACITY FACTOR
2022	99,95	100	96,7
2021	98.3	99.6	95.2
2020	99.8	99.8	96.6





TEKNOLOGI KONVERSI LAINNYA



Single Flash Steam



Double Flash Steam



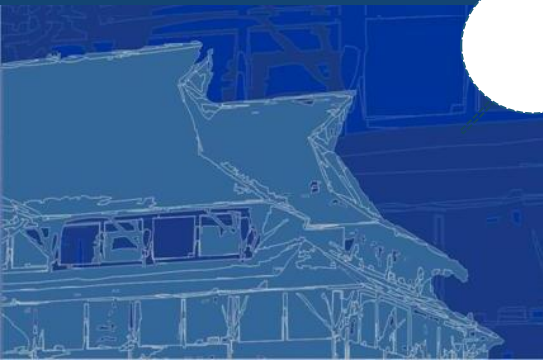
Triple Flash Cycle



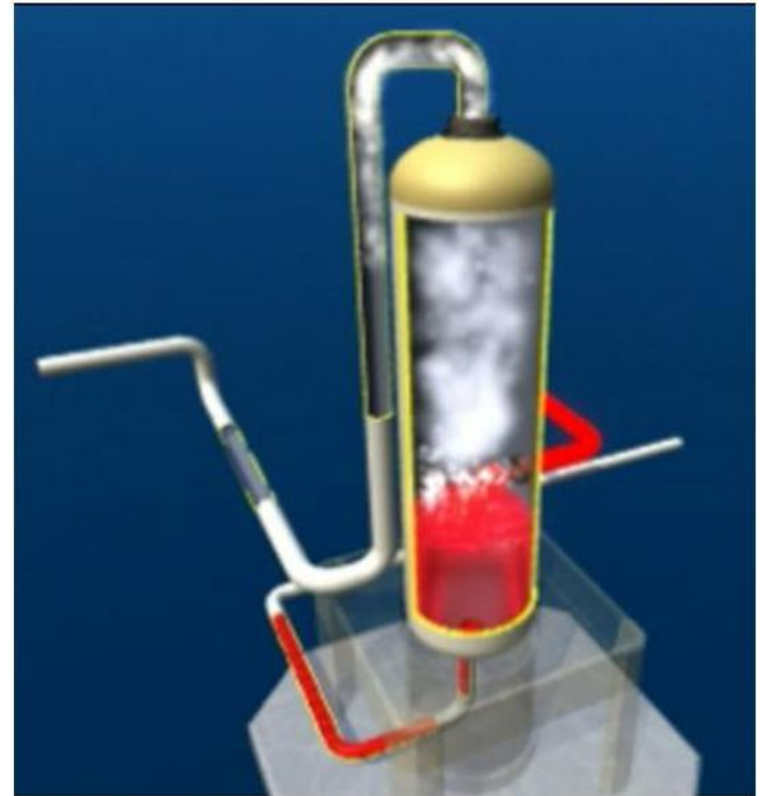
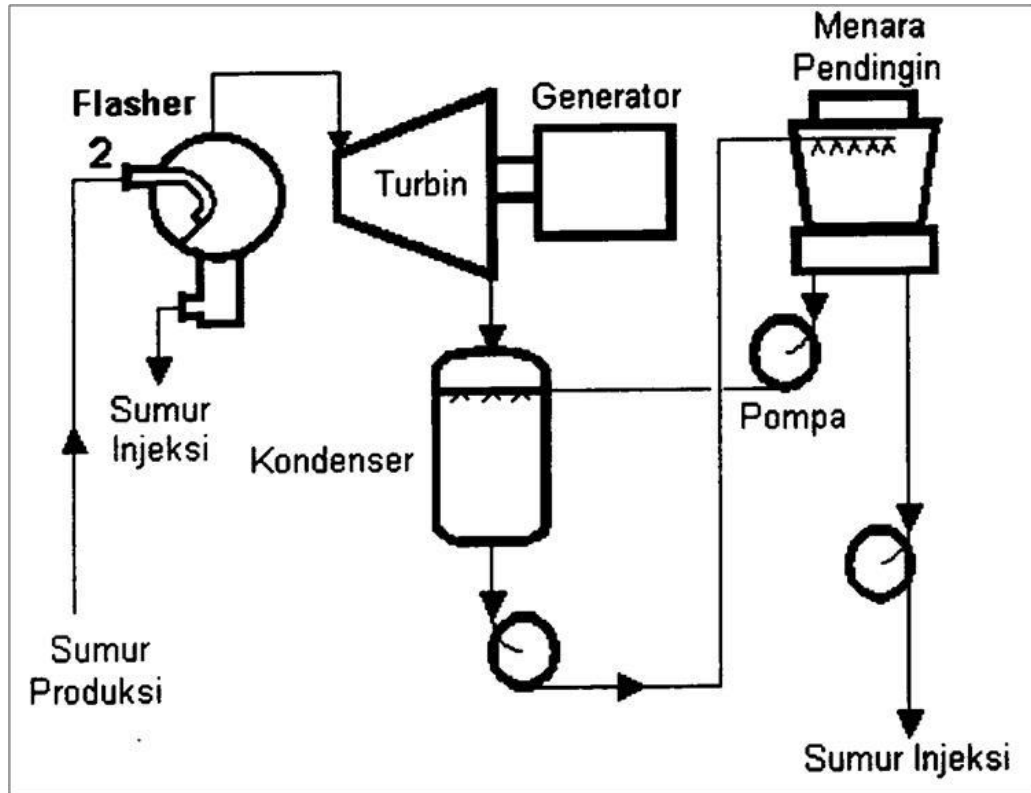
Combined Cycle



Well Head Generating Unit



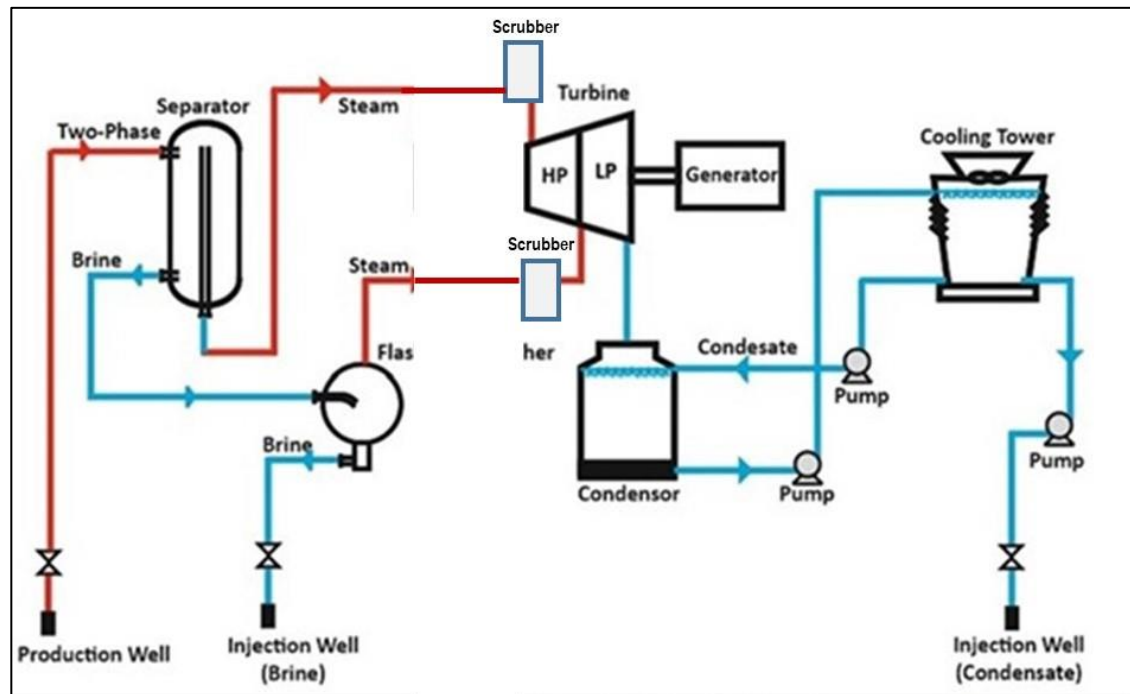
Single Flash Steam Cycle (Siklus Uap Hasil Penguapan)



Sistem pembangkit listrik siklus uap hasil penguapan atau *single flash steam* digunakan bilamana fluida dikepala sumur dalam kondisi air jenuh (saturated liquid). Fluida dialirkan ke sebuah *flasher* agar menguap.

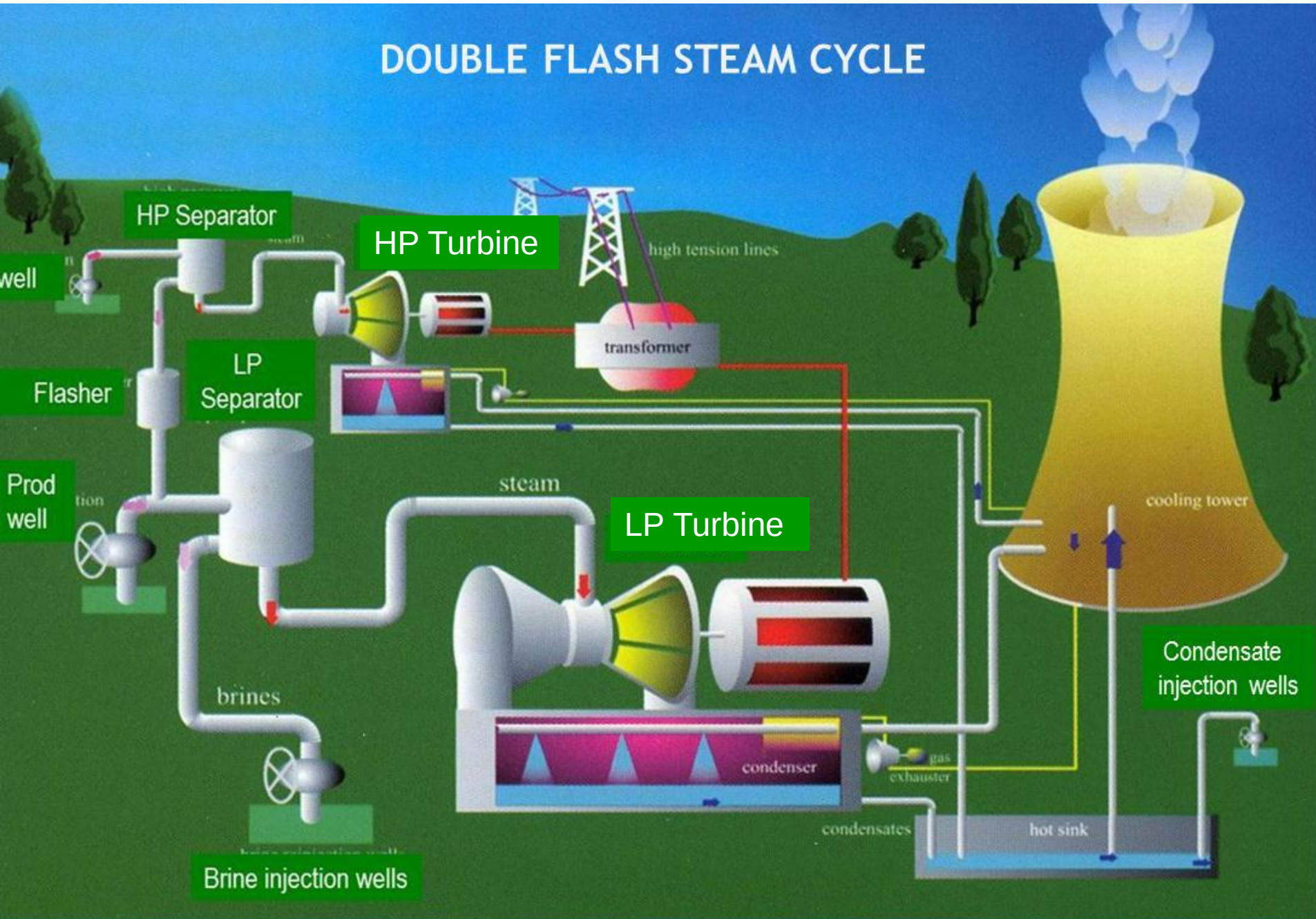
DOUBLE FLASH STEAM CYCLE

Uap yang digunakan adalah uap dari hasil pemisahan fluida dalam separator dan uap dari *flasher* yang merupakan hasil penguapan air yang keluar dari separator



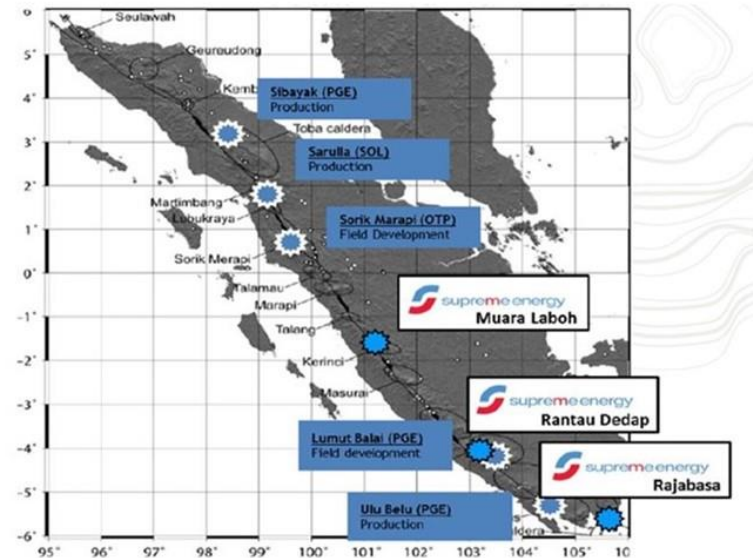
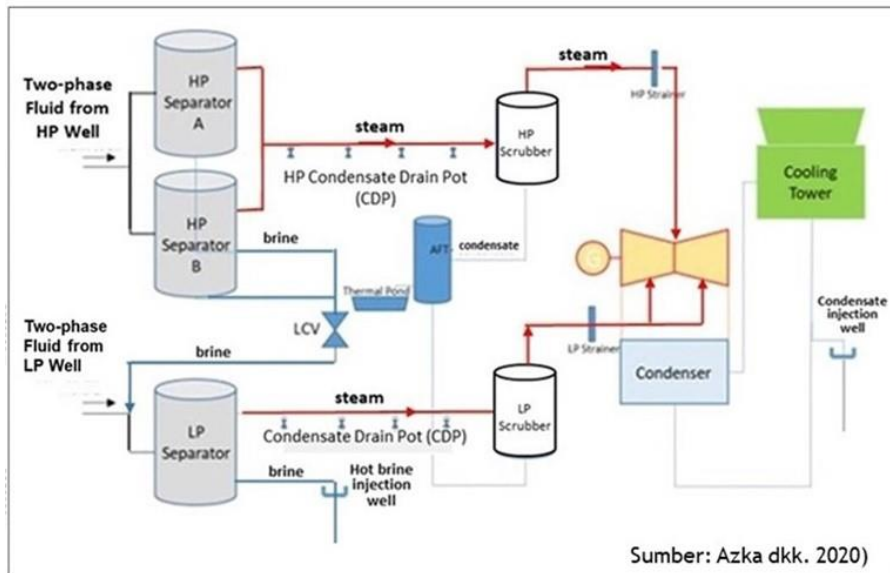
Pemakaian di lapangan Hatchobaru (Jepang), dan Krafla (Iceland)., Muara Laboh Unit-1: 85 MW (2019) & Rantau Dedap Unit-1 : 91,2 MW (2021)

DOUBLE FLASH STEAM CYCLE



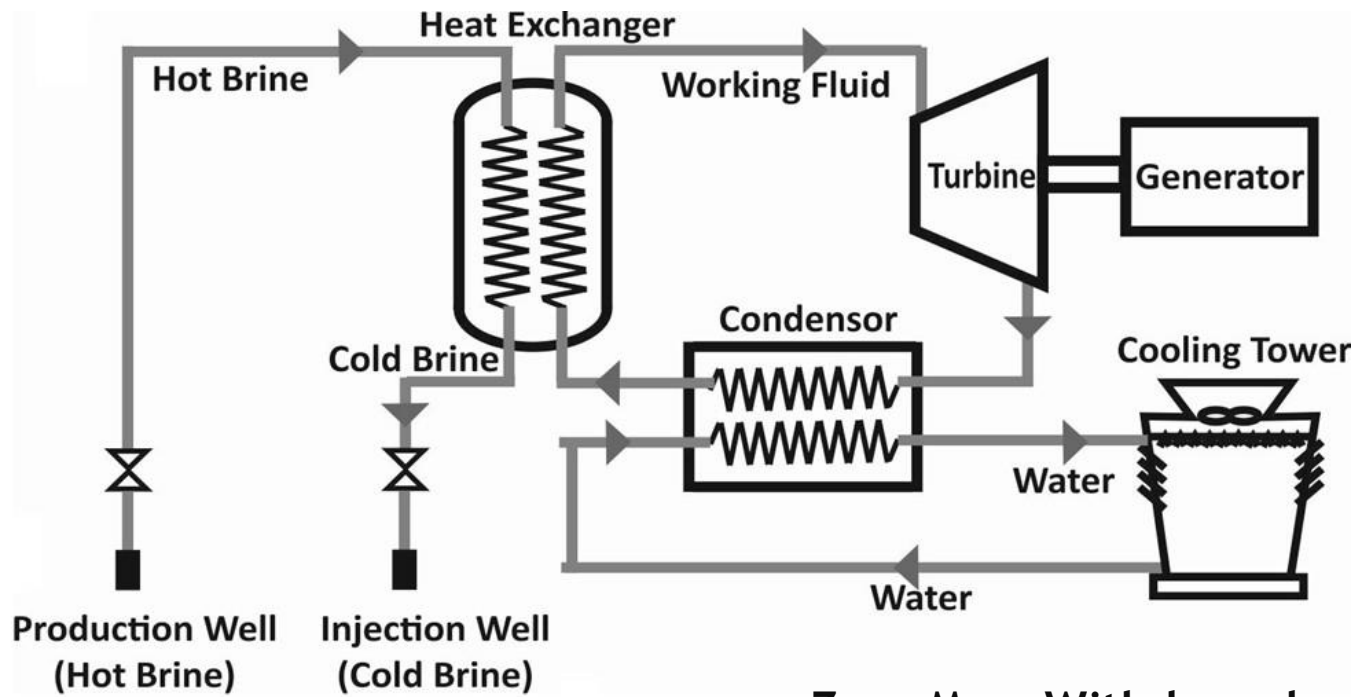
Schematic diagram of geothermal utilization for electricity

TEKNOLOGI KONVERSI DOUBLE FLASH DI LAPANGAN MUARA LABOH

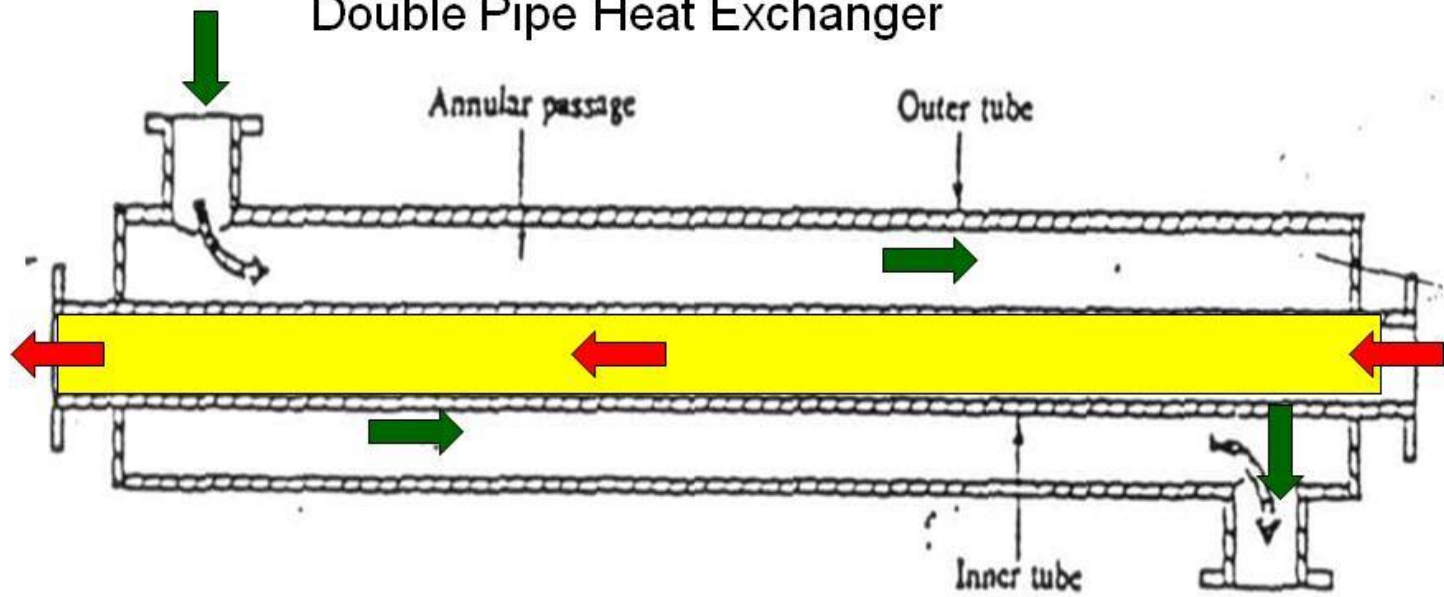


SIKLUS BINARI (BINARY CYCLE)

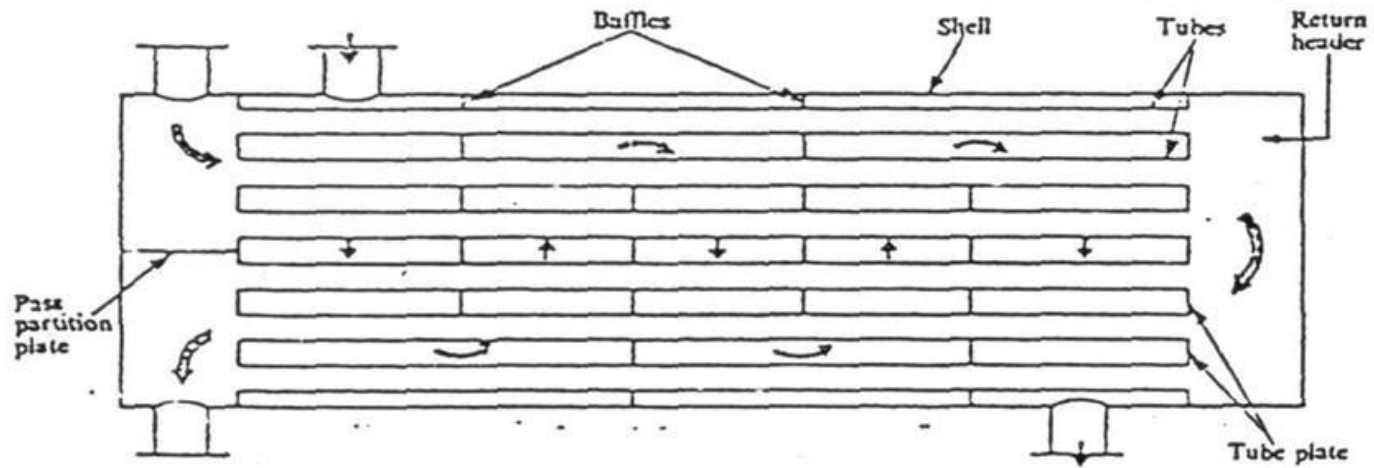
Dalam siklus pembangkit ini fluida sekunder (*isobutane, isopentane or ammonia*) dipanasi oleh fluida panas bumi melalui mesin penukar kalor atau *heat exchanger*. Fluida sekunder menguap pada temperatur lebih rendah dari temperatur titik didih air pada tekanan yang sama. Fluida sekunder mengalir ke turbin dan setelah dimanfaatkan dikondensasikan sebelum dipanaskan kembali oleh fluida panas bumi.

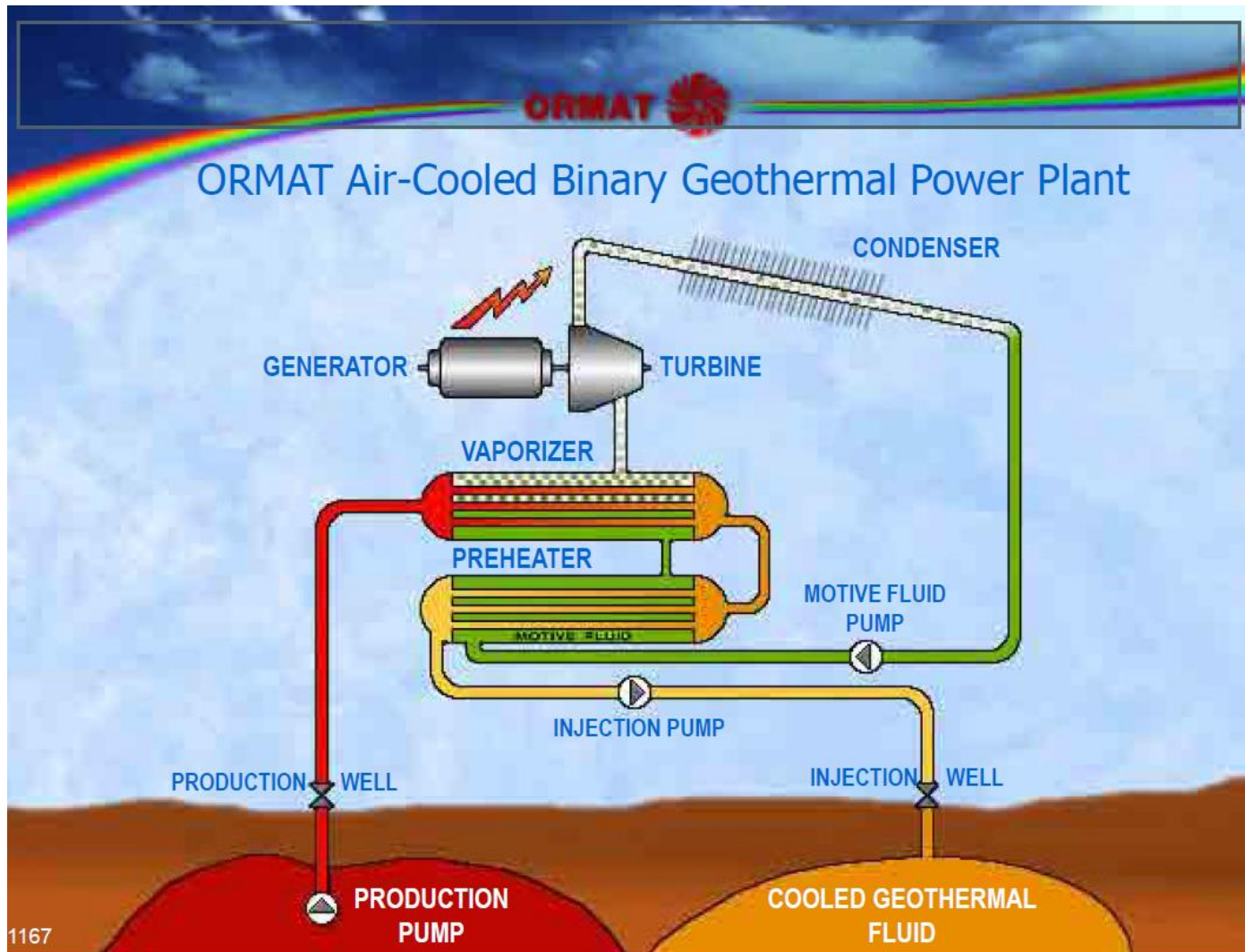


Double Pipe Heat Exchanger



Shell and Tube Heat Exchanger

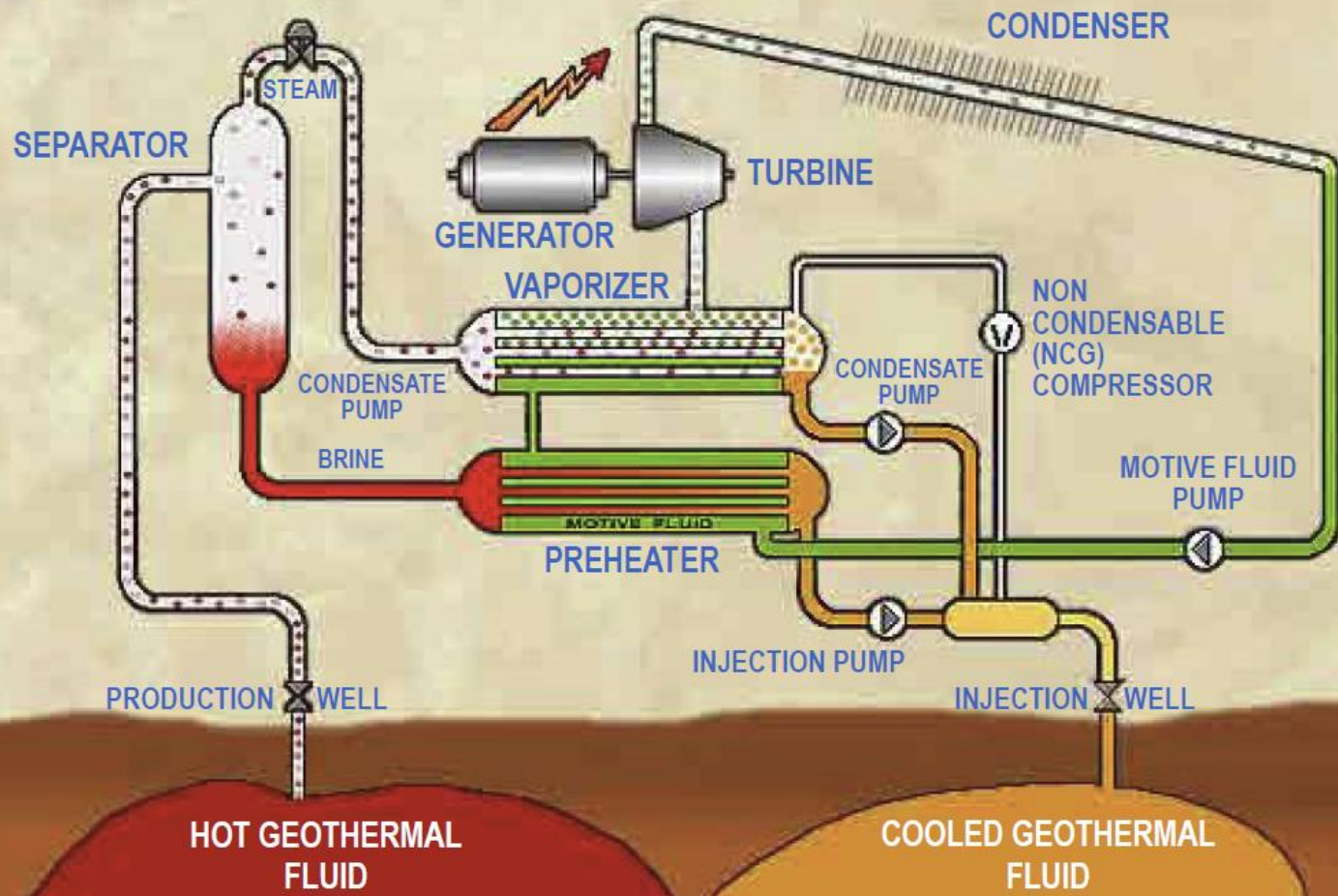




Simple binary cycle OEC's in several variations for low enthalpy fluid and separated brine (or steam only for small scale)

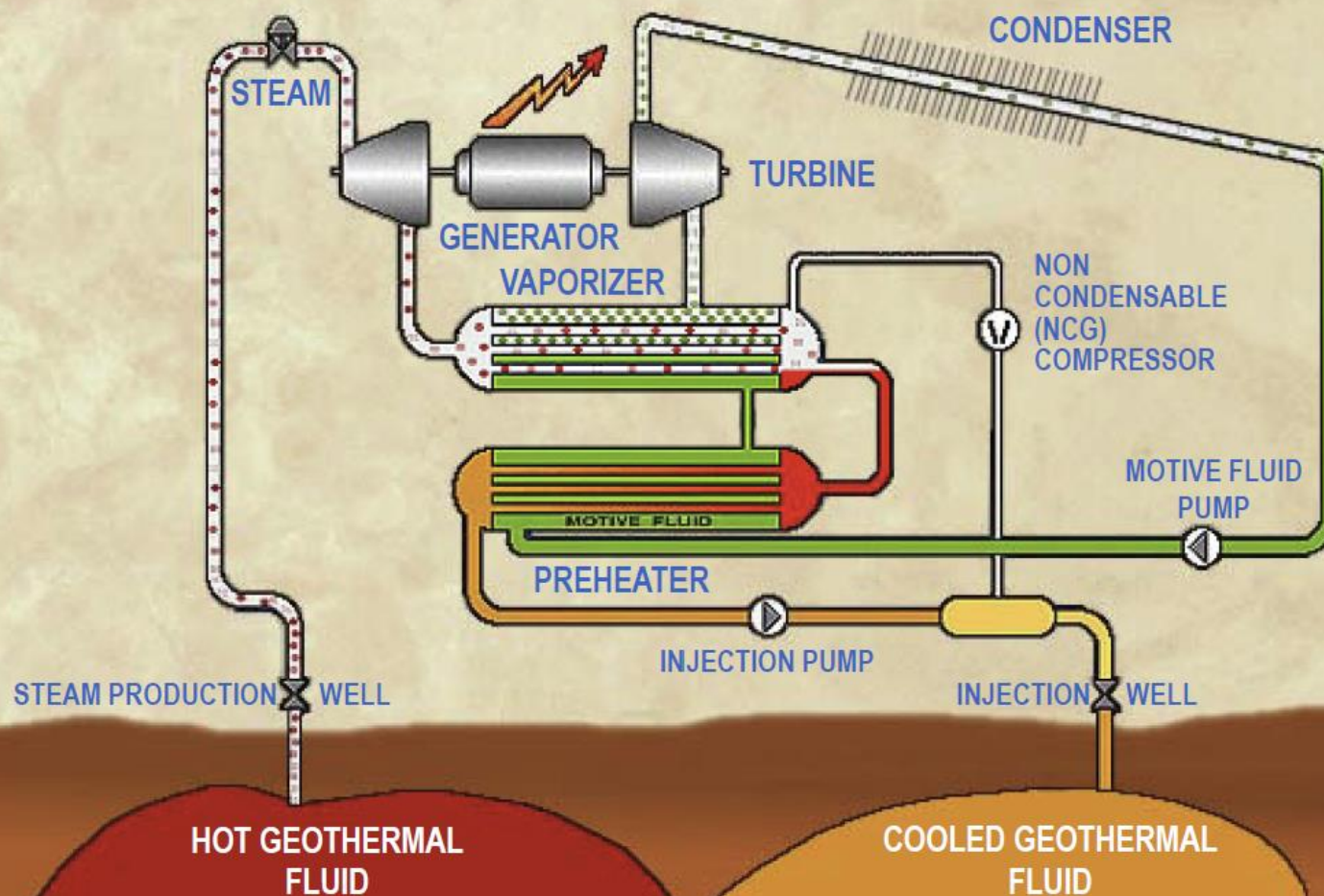


ORMAT Two-Phase Binary Geothermal Power Plant

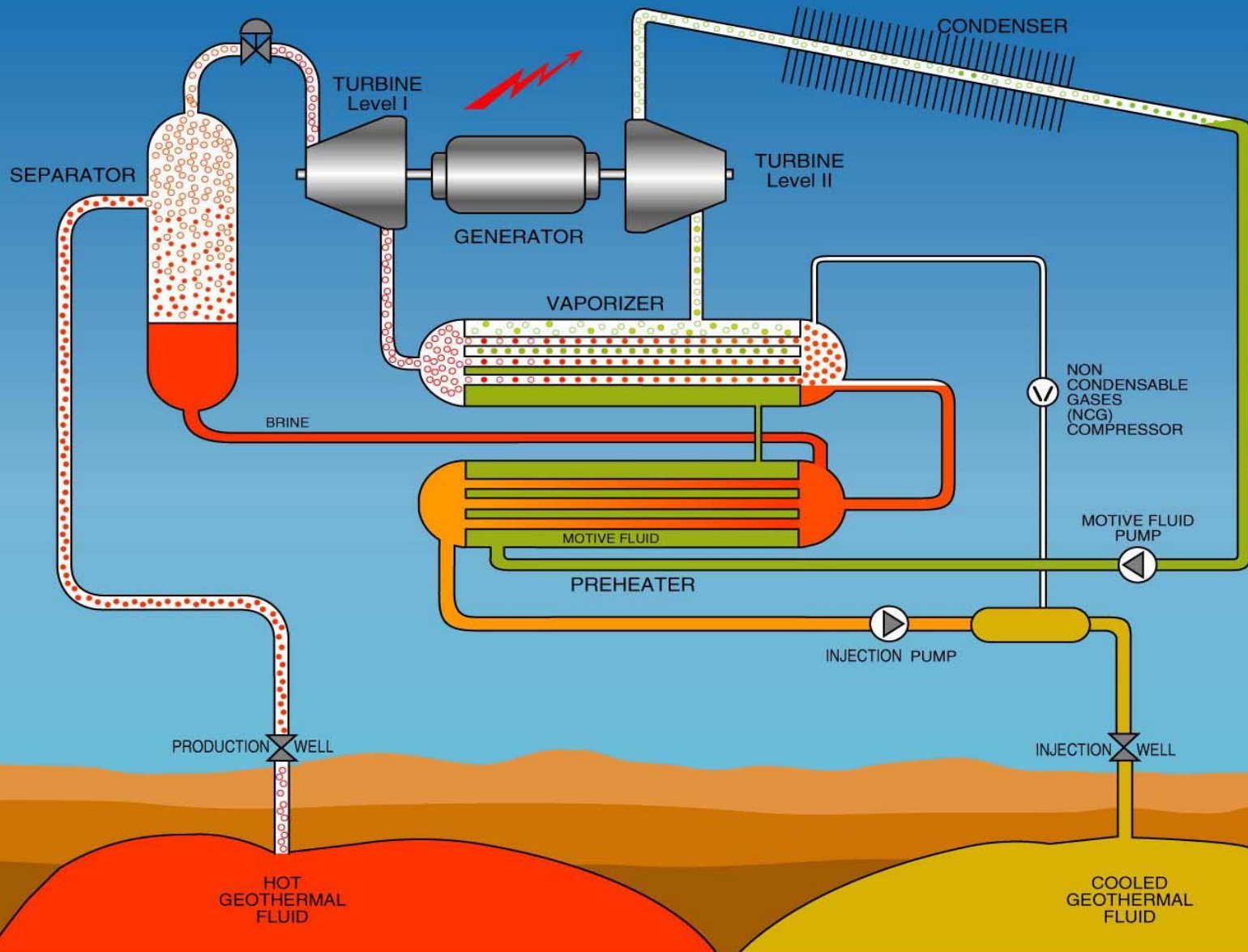


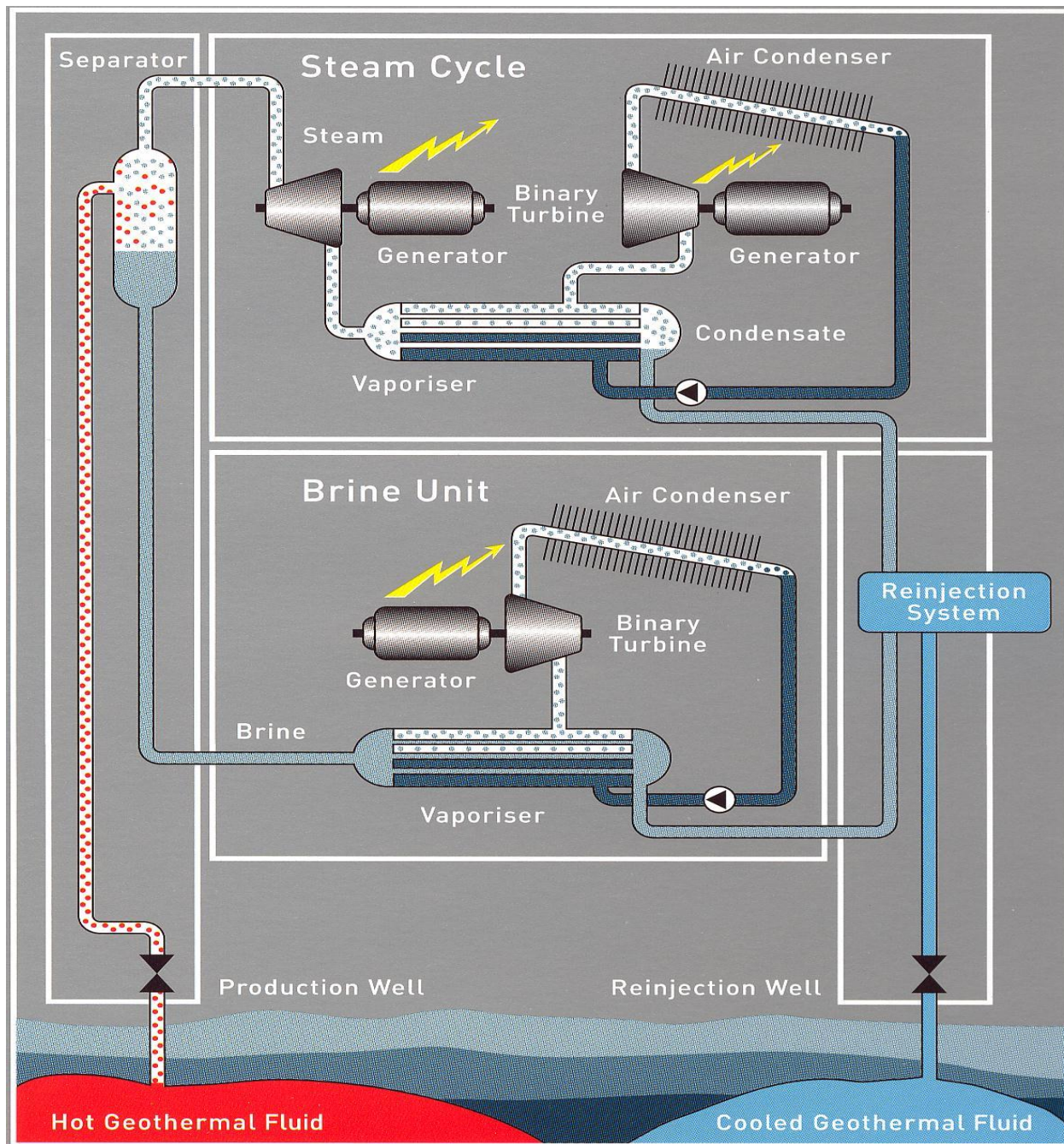


ORMAT® Geothermal Combined Cycle Power Plant



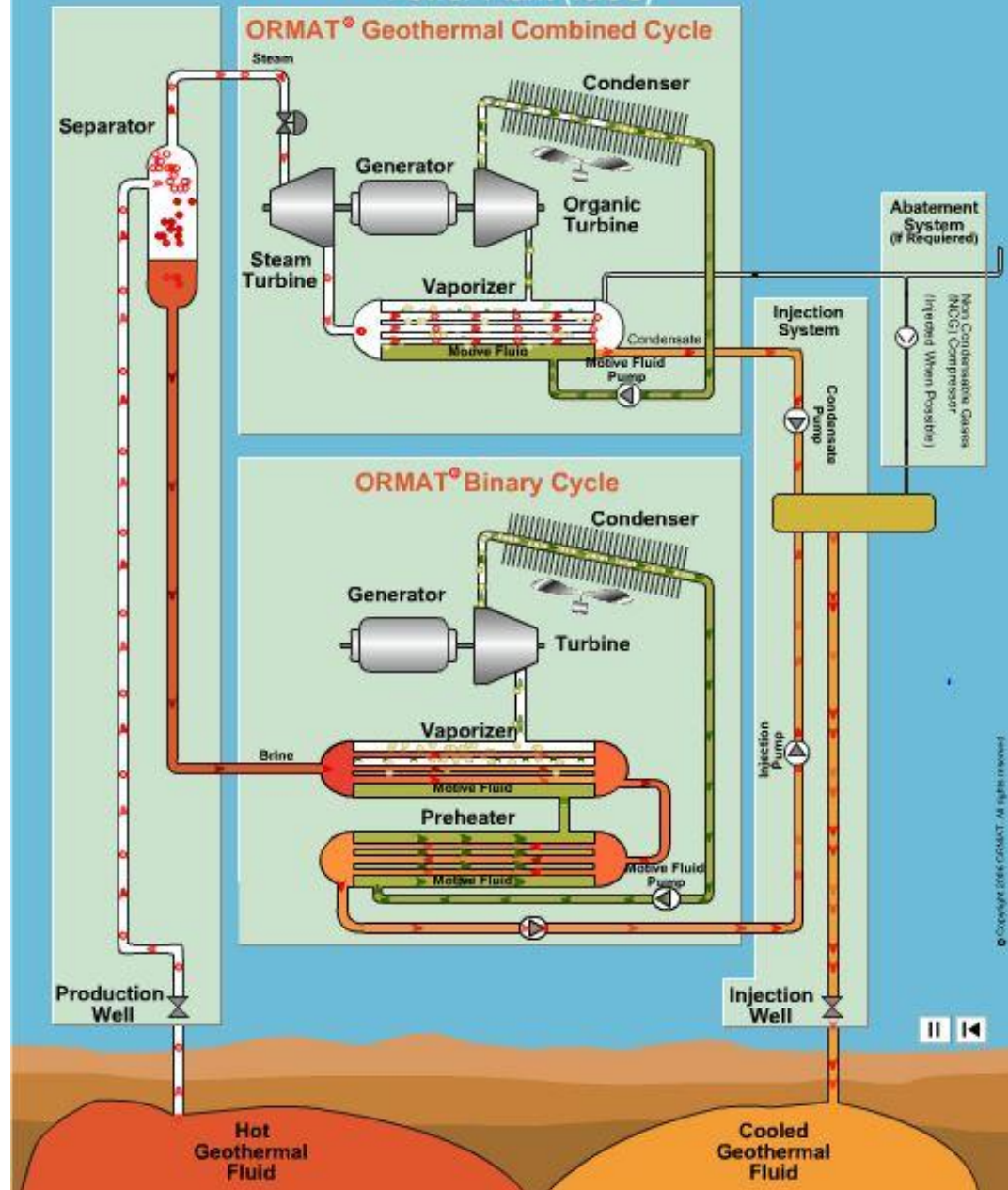
Geothermal Combined Cycle Power Plant







Integrated Geothermal Combined Cycle Power Plant (IGCC)

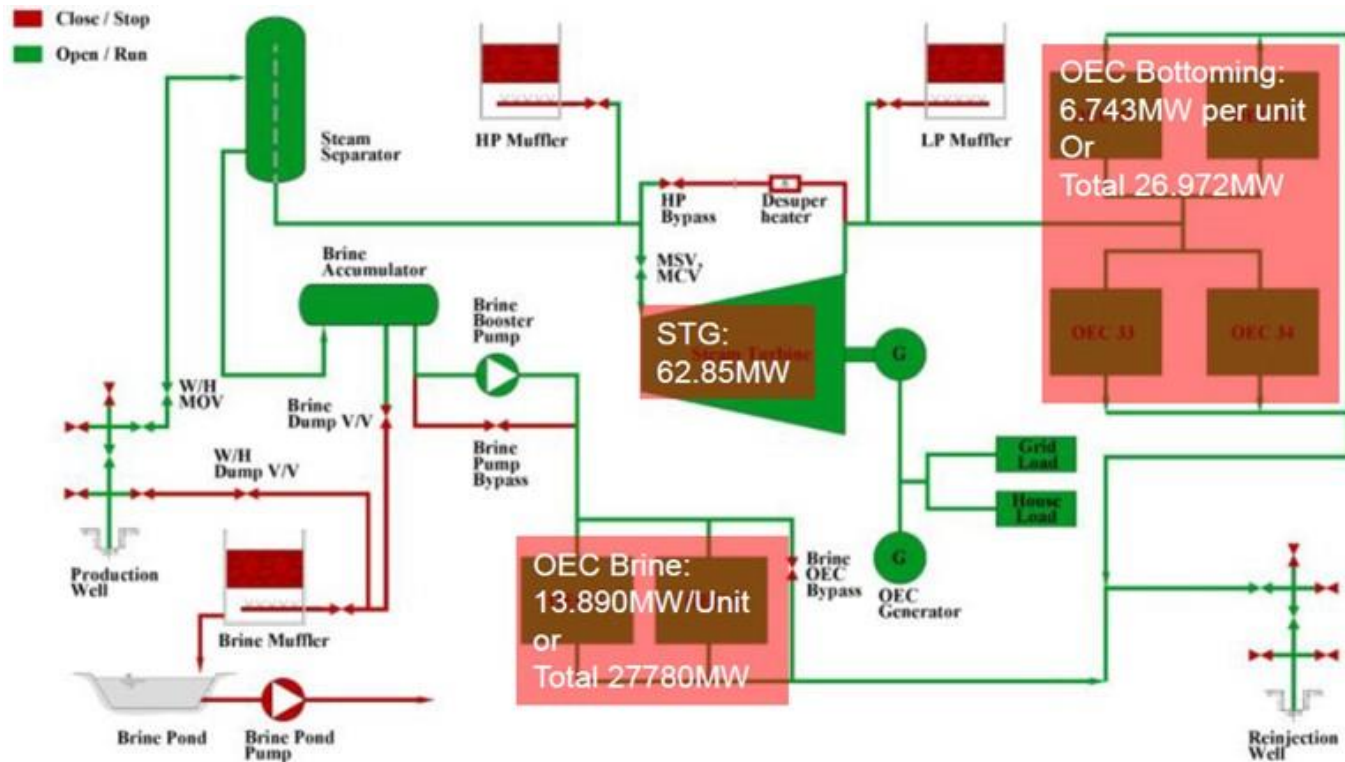


Cycles

- SOL is operating 3x110MW Geothermal Power Plant with Integrated Geothermal Combined Cycles
 - 1 x 110MW Silangkitang (SIL-1)
 - 2 x 110MW Namora I Langit (NIL-1& NIL-2)
- Integrated Geothermal Combined Cycles Unit (IGCCU) consist of:
 - Single Flash – Backpressure STG
 - Binary:
 - OEC Bottoming
 - OEC Brine

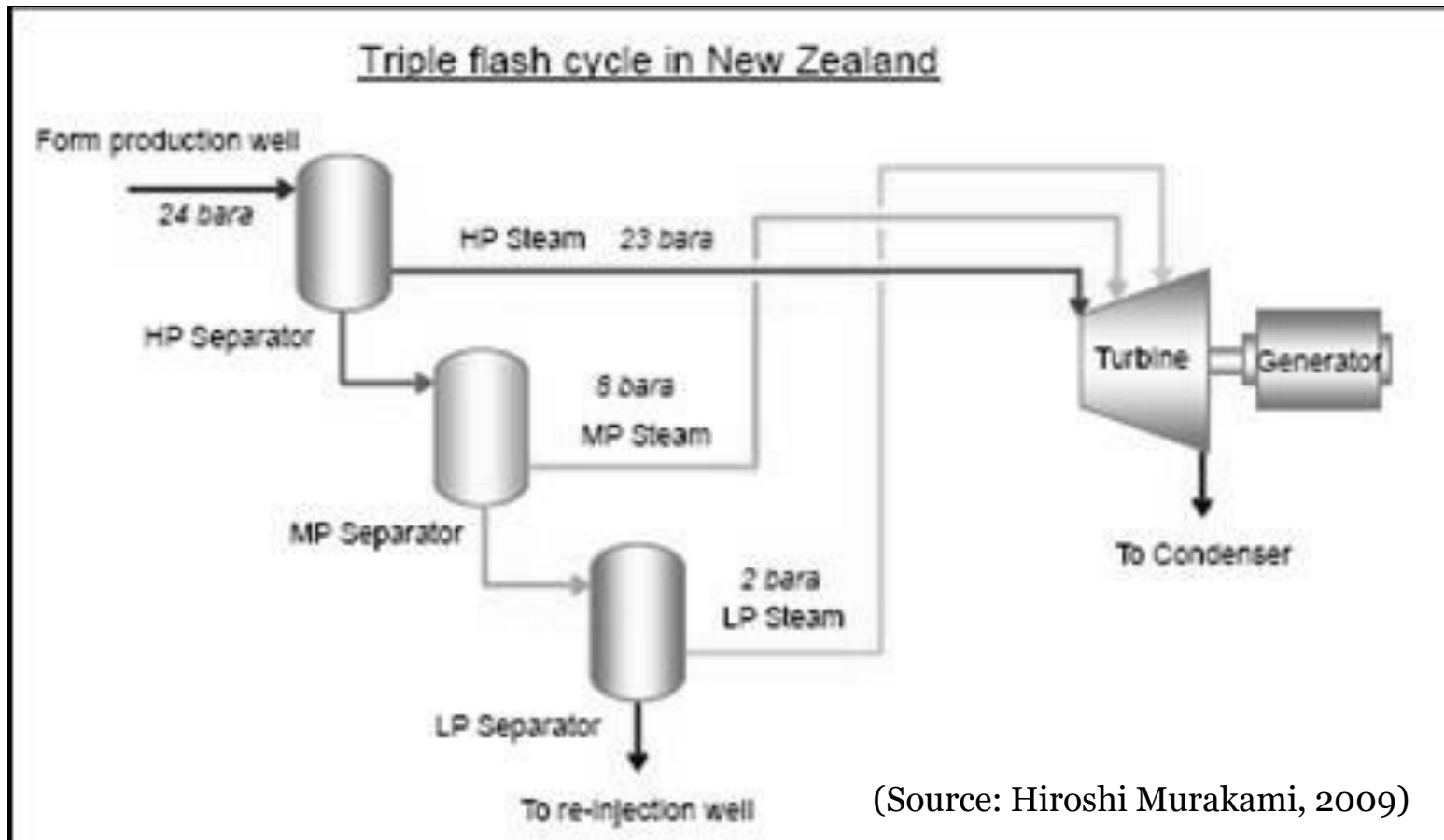
Note: OEC = ORMAT Energy Converter

INTEGRATED GEOTHERMAL COMBINED CYCLE UNITS POWER PLANT (ORMAT) DI SARULLA



- Uap dari separator kemudian dialirkan ke *Steam Turbine Generator (STG)* dan menghasilkan listrik berkapasitas 62.85 MW, kemudian uap yang keluar dari turbin dialirkan ke *Binary Unit: ORMAT Energy Converter (OEC) Bottoming*,
- Uap dari dari *N-Pentane* kemudian dialirkan ke turbin dan menghasilkan listrik 6.743 MW/unit. Pembangkit ini terdiri dari 4 unit, sehingga total listrik yang dibangkitkan 26.972 MW.
- Air dari separator dialirkan ke binary unit *ORMAT Energy Converter (OEC) Brine*. Dalam siklus ini terdapat 2 unit, menghasilkan 13.890 MW/unit, sehingga total listrik yang dibangkitkan adalah 27.780 MW.

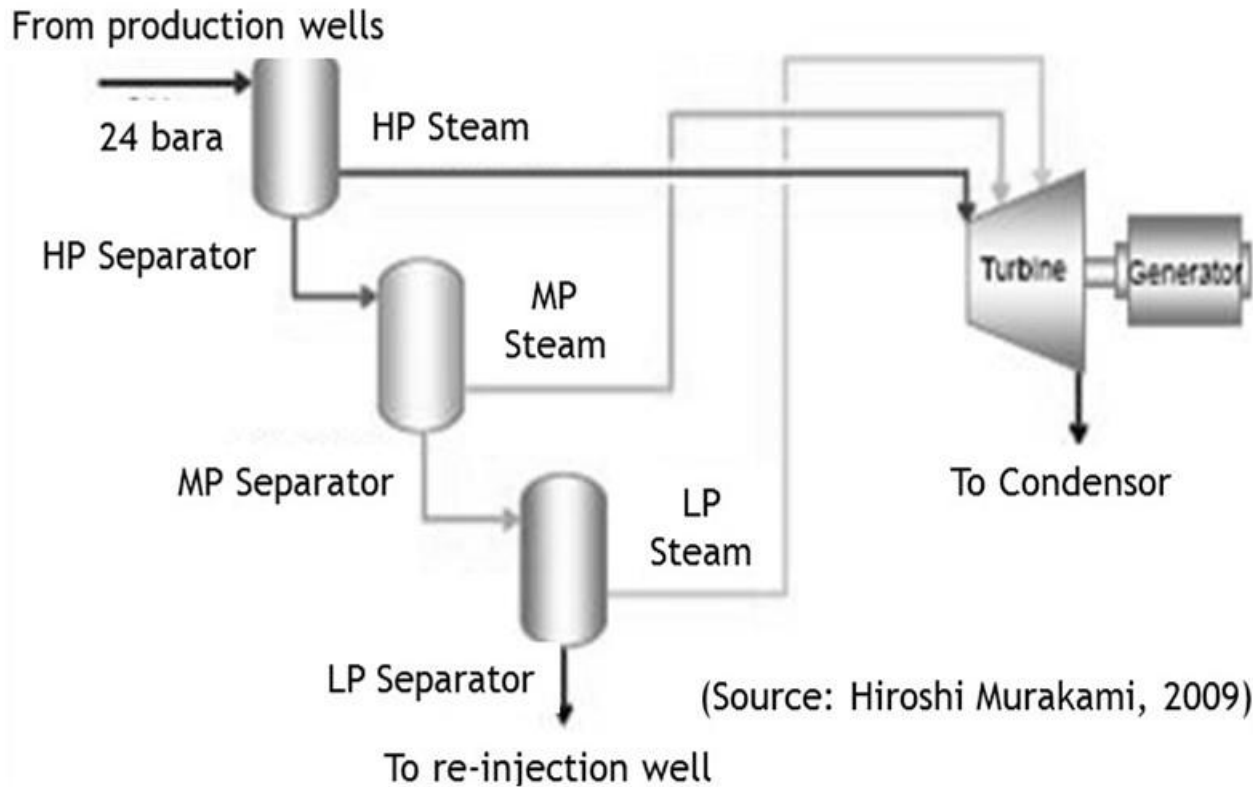
Siklus Pemisahan Tiga Tingkat (Triple Flash Cycle) di New Zealand



SIKLUS PEMISAHAN TIGA TINGKAT (TRIPLE FLASH CYCLE) DI NEW ZEALAND

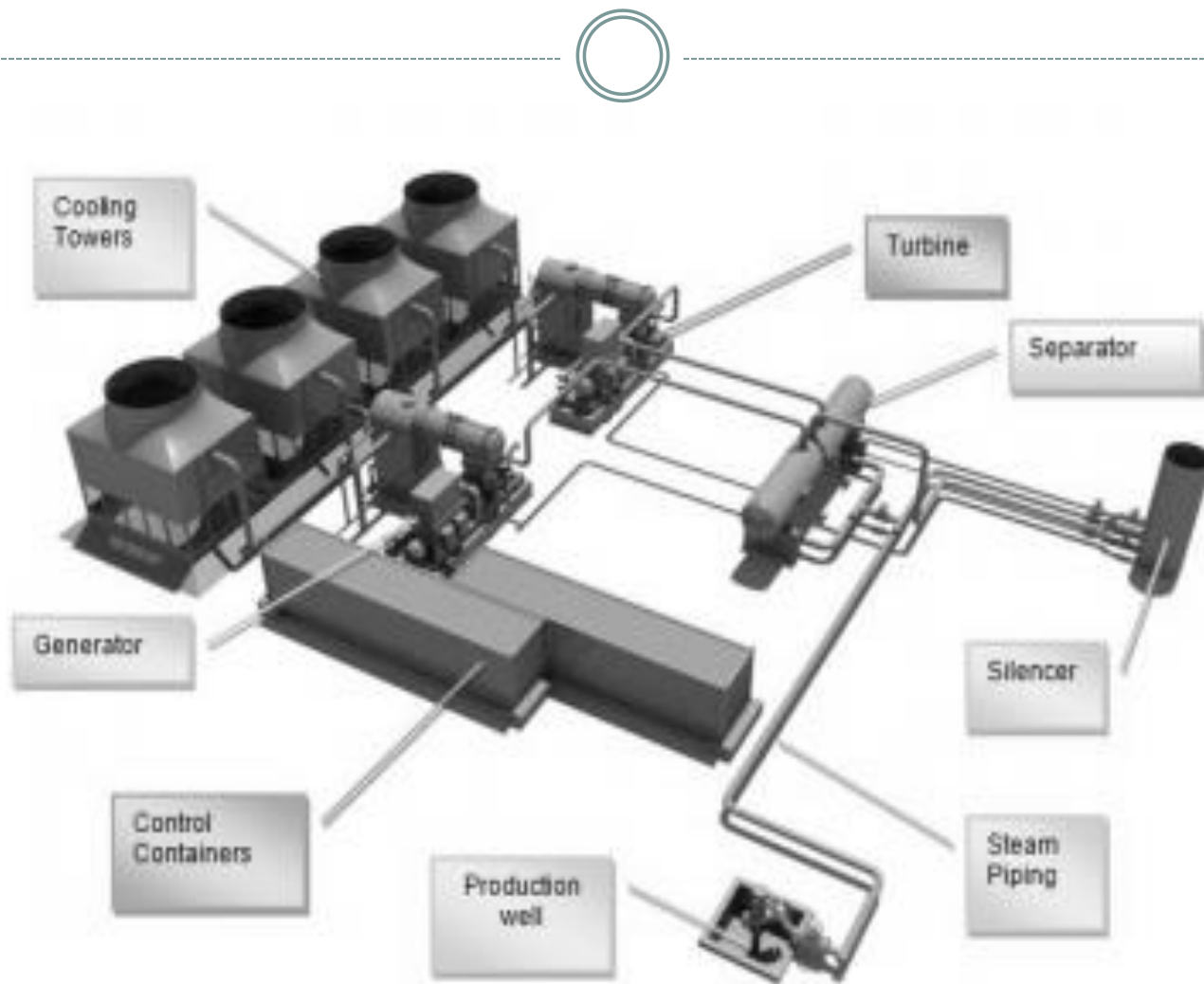


Triple Flash Cycle in New Zealand

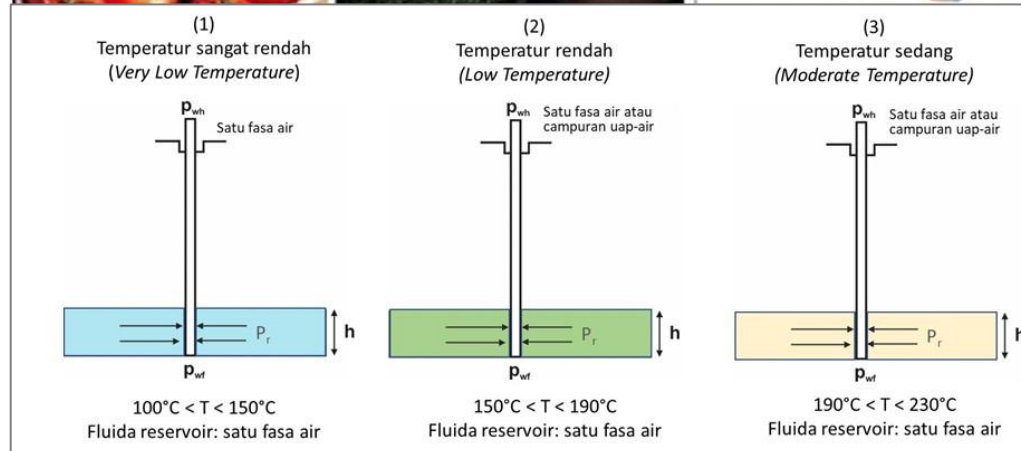


Power Plant	Nga Awa Purua
Output	140 MW
Inlet Steam Pressure	HP 23.5 bara, IP 8.4 bara, LP 2.3 bara
Inlet Steam Temp.	HP 221°C, IP 172°C, LP 125°C
Exhaust Pressure	0.085 bara
Steam Flow	HP 617 t/h, IP 113 t/h, LP 103 t/h

WELLHEAD GENERATING UNIT

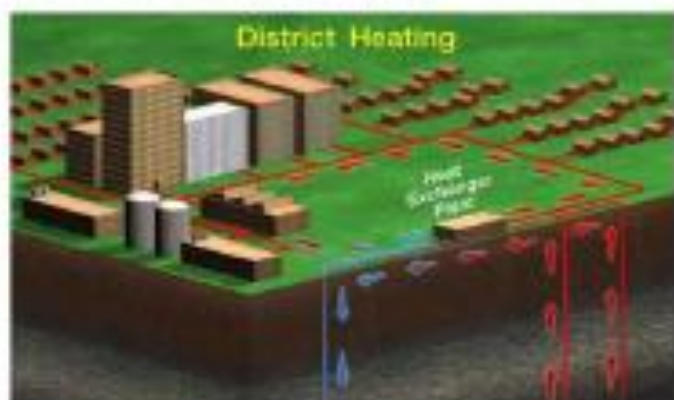


NON-ELECTRICITY UTILIZATION OF GEOTHERMAL ENERGY (DIRECT USE)



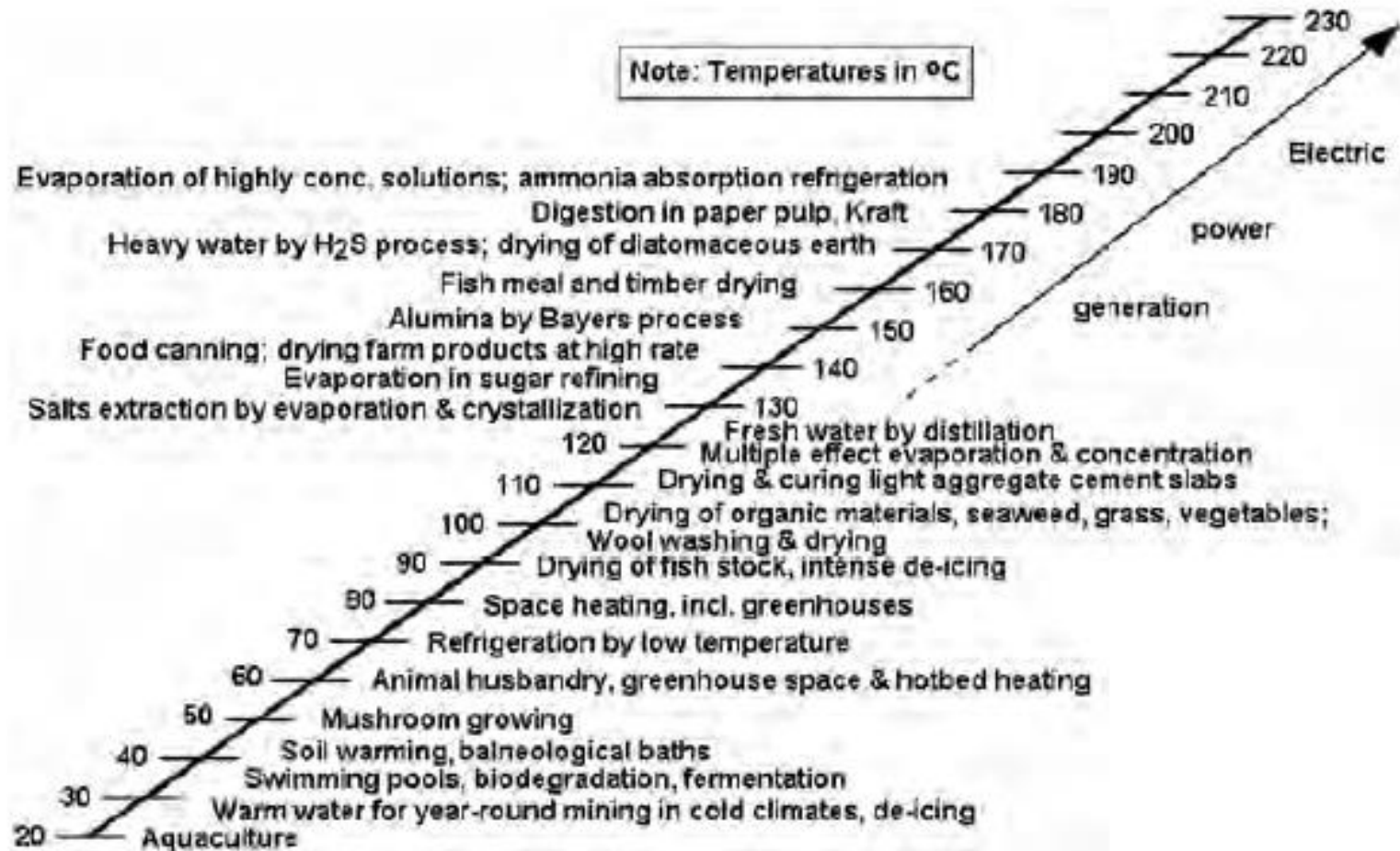
PEMANFAATAN LANGSUNG

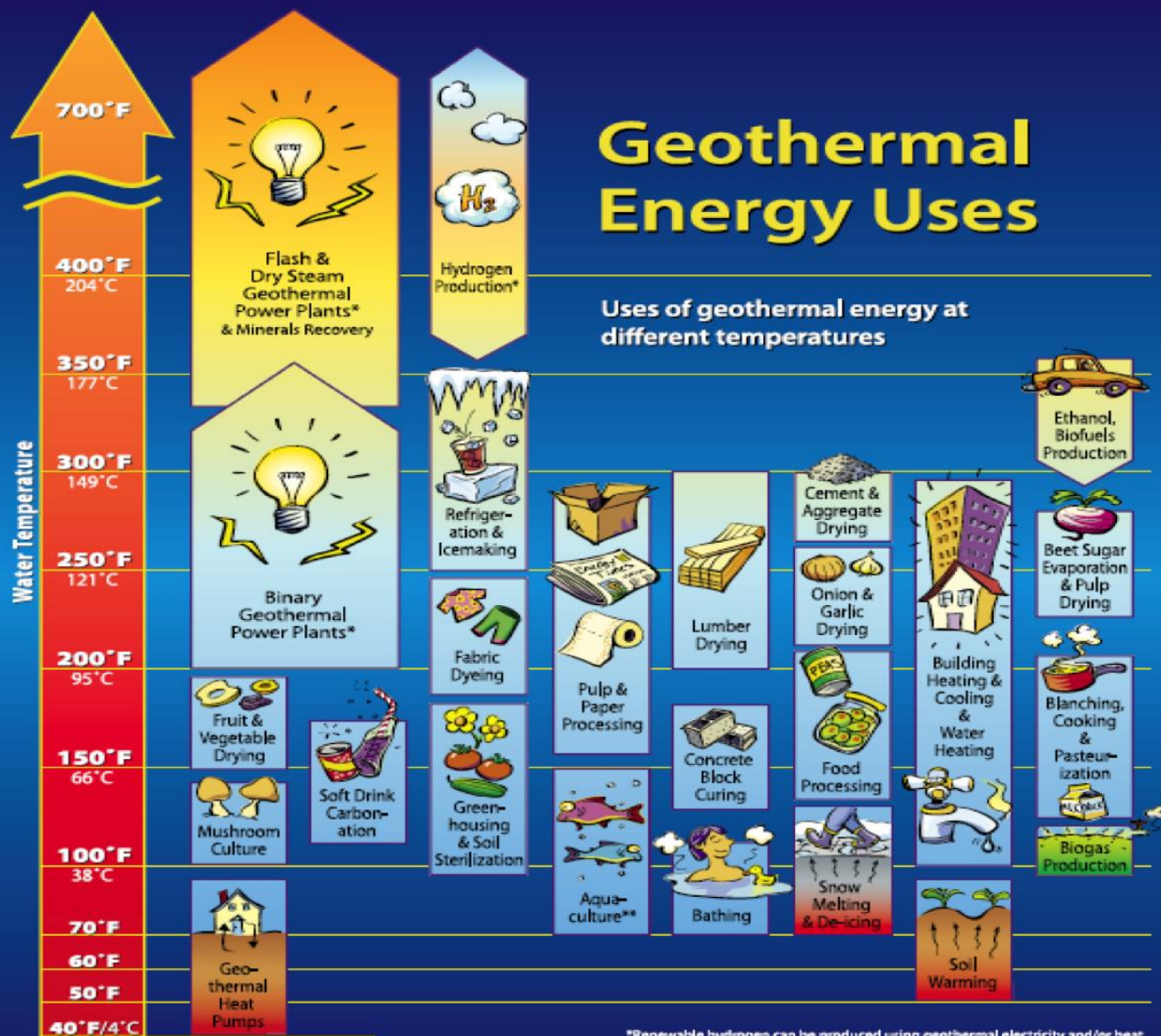
- Fluida panas bumi dimanfaatkan untuk sektor non-listrik di 72 negara, antara lain untuk pemanasan ruangan, pemanasan air, pemanasan rumah kaca, pengeringan hasil produk pertanian, pemanasan tanah, pengeringan kayu, kertas dll.



PEMANFAATAN LANGSUNG (DIRECT USE) DI BEBERAPA NEGARA

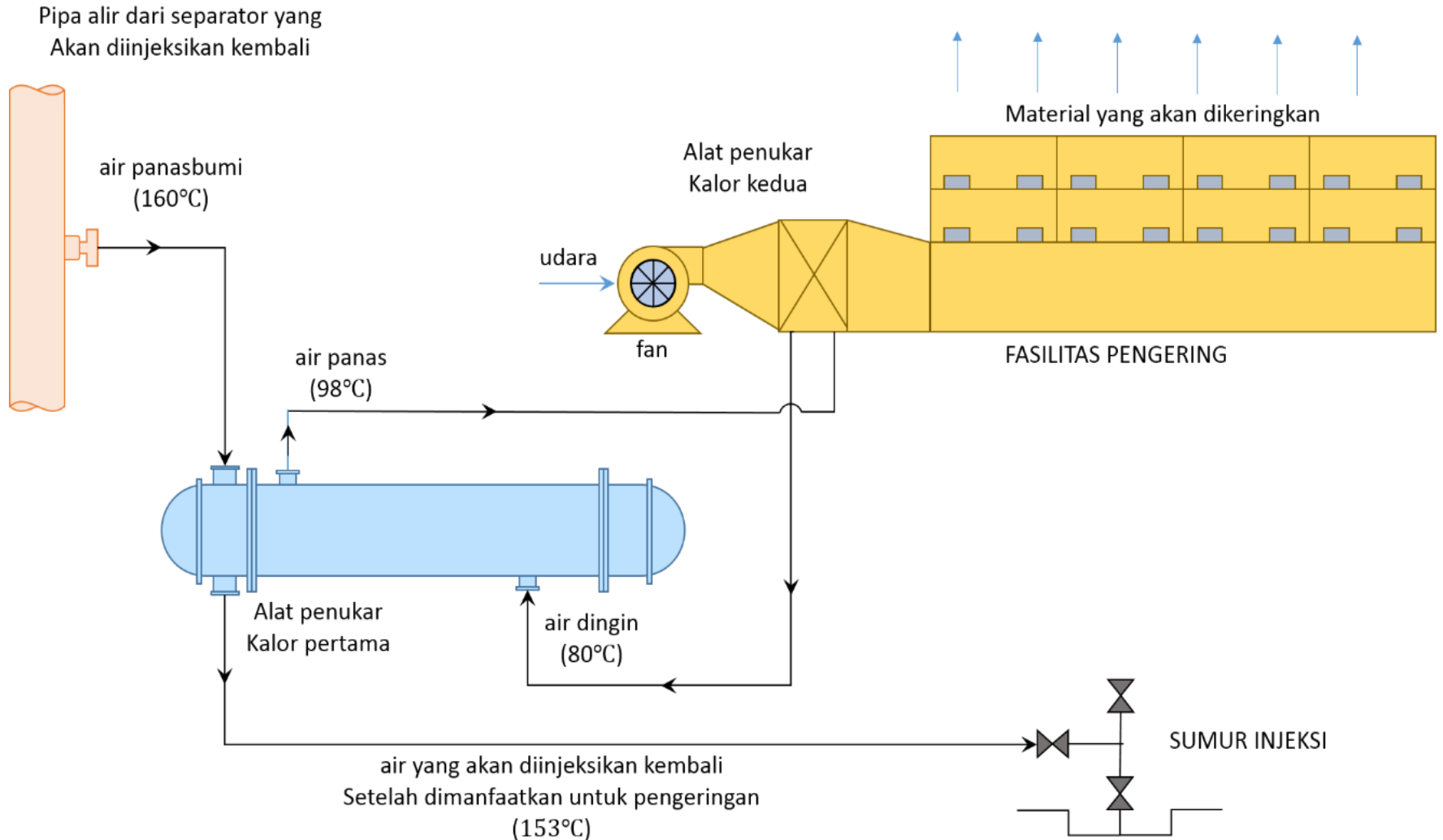
Lindal Diagram



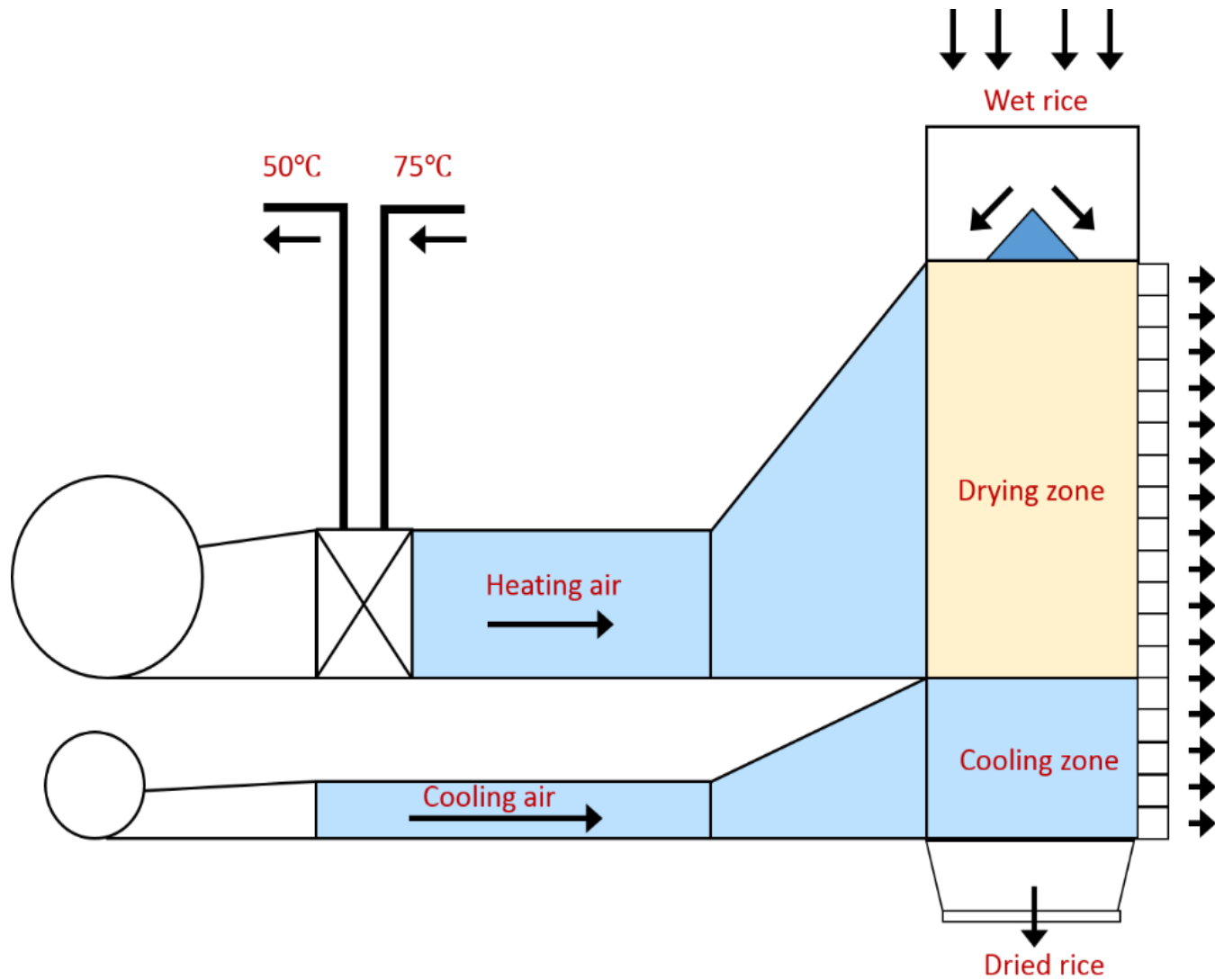


NON-ELECTRICITY UTILIZATION OF GEOTHERMAL FLUID

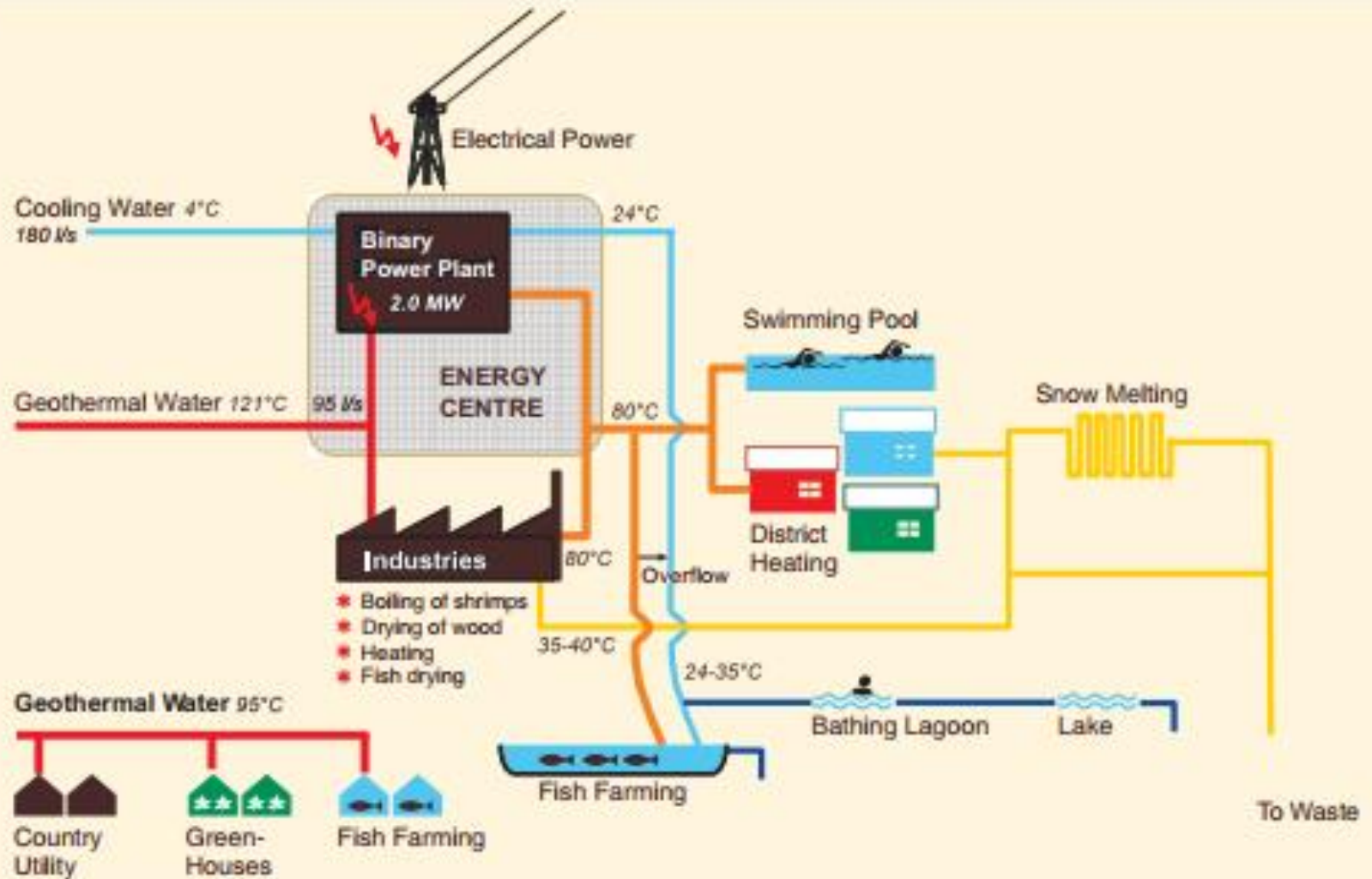
Pilot Project di Philiipina



UTILIZATION OF GEOTHERMAL FLUID FOR RICE DRYING IN MACEDONIA



Idealized Diagram Showing Multiple Use of Geothermal Energy



Source | EOR (Island Geosurvey), 2005, direct communication.

PEMANFATAN LANGSUNG (*DIRECT USE*) DI INDONESIA



Peluang usaha panas bumi tidak hanya terbatas pada pengembangan pembangkit listrik panas bumi (pemanfaatan tidak langsung), tetapi juga untuk pemanfaatan langsung seperti pemanasan, pariwisata, dan sebagainya. Peluang lain adalah melalui pemanfaatan potensi panas bumi bersuhu rendah, pembangkit skala kecil dan jasa perusahaan untuk mendukung bisnis inti panas bumi.

PEMANFAATAN LANGSUNG FLUIDA PANAS BUMI DI INDONESIA



Contoh: Di lapangan panas bumi Lahendong – Sulawesi Utara



Photo: Nenny Saptadji, 2014

- ❖ Fluida panas bumi digunakan untuk masak nira jadi gula aren. Gula aren semua di ekspor. Produksi sekitar 250 kg/hari. Dulu dg kayu bakar, utk menghasilkan satu kualiti/penggorengan gula aren butuh waktu 3-4 jam tetapi dg fluida panas bumi hanya butuh 30 menit.
- ❖ Fluida panas bumi digunakan untuk pengeringan kopra.
- ❖ Area manifestasi dijadikan area wisata.
- ❖ Akan membuat "Geopark"



NON-ELECTRICITY UTILIZATION OF GEOTHERMAL

Di Kamojang



Di Lahendong



Pemanfaatan Fluida Geotermal untuk Destilasi Akar Wangi Menjadi Minyak Astiri



The diagram illustrates a steam heating system for a Separator. The process involves two main fluid loops: a heating medium loop (blue) and a process fluid loop (red).

Heating Medium Loop (Blue):

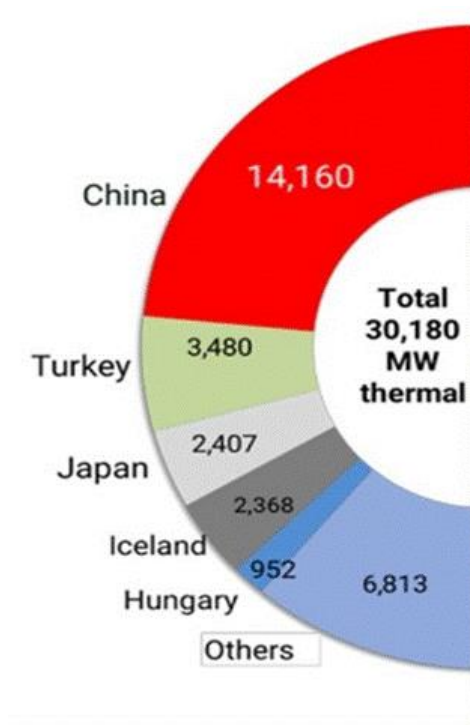
- Tanki-01:** The source of heating medium, with $m_b = 86,2 \text{ kg/s}$ and $T = 148,9^\circ\text{C}$.
- Pompa-02:** A pump that circulates the heating medium from Tanki-01.
- Heat Exchanger:** The heating medium flows through a horizontal heat exchanger, heating the process fluid.
- Aplikasi I, Aplikasi II, Aplikasi III:** The heated heating medium is distributed to three application points.
- Return:** The cooled heating medium returns to Tanki-01.

Process Fluid Loop (Red):

- Separator:** The process fluid enters the Separator at $m_b = 86,2 \text{ kg/s}$ and $T = 182^\circ\text{C}$.
- Heating:** The process fluid is heated by the heating medium in the heat exchanger. The outlet temperature is $149,1^\circ\text{C}$ with $m_b = 86,2 \text{ kg/s}$.
- Pompa-01:** A pump that circulates the process fluid from the heat exchanger to the Separator.
- Control:** A valve labeled "NO" is located on the line between the heat exchanger and the Separator.
- Discharge:** The heated process fluid is discharged from the Separator at $m_b = 3,33 \text{ kg/s}$ and $T = 171^\circ\text{C}$.

GEOHERMAL DIRECT USE WORLDWIDE AND INDONESIA

WORLDWIDE



INDONESIA : Rank 74 in the world



- Bathing, tourism, mushroom farming, tea drying, palm sugar & coffee processing, catfish farming.
- Potential Opportunities: Silica Extraction & Lithium Production



Terimakasih